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PROJECT REPORT

ON

"Detection of Fraudulent Sellers in Online Marketplaces"

SUBMITTED IN PAFRTIAL FULFILMENT OF THE REQUIREMENT FOR THR AWARD OF THE DEGREE OF

BACHELOR OF COMPUTER APPLICATIONS

OF

BANGALORE NORTH UNIVERSITY

Submitted by

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NEW HORIZON COLLEGE
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Accredited by NAAC with 'A' Grade

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CERTIFICATE

This is to certify that **SIDDHANT YOJIT (Reg No: R2014693)** is a bonafide student of New Horizon college and has carried out a project entitled "**Detection of Fraudulent Sellers in Online Marketplaces**" under the guidance of **Ms. Apoorva**. This project report has been submitted during the academic year 2022-2023 in a partial fulfilment of requirements of the Degree in Bachelor of Computer Application, Bangalore North University.

Ms. Apoorva **Dr. Nagaraju**

Assistant Professor **head of the department**

Guide

Dr. R. Bodhisatvan.

Principal

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DELARATION

I **Siddhant Yojit (Reg No: R2014693)** do here by declare that the project work entitled– "**Detection of Fraudulent Sellers in Online Marketplaces**" is a bonafide work carried out by me under the guidance of **Ms. Apoorva**.This project is my original work and has not been presented for any other university. This project has been submitted as part fulfilment of requirements for the Degree of Bachelor of Computer Application, Bangalore North University.

SIDDHANT YOJIT

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Date:

Place: NHC-M, Bangalore

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I sincerely thank the Chairman, **Dr Mohan Manghnani** of New Horizon College for providing me the necessary facilities for the completion of the project.

With great pleasure I place on record my deep gratitude to the principal,

Dr R. Bodhisatvan. for providing all the facilities for the completion of the project.

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SIDDHANT YOJIT

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CONTENTS

SL.NO	SUBJECT	PG.NO
1.	INTRODUCTION A. PROJECT DESCRIPTION B. FRAUD PRODUCT DETECTION C. WEB PERSONALISATION D. PROJECT PURPOSE E.	6-7
2.	ABSTRACT	8
3.	HARDWARE AND SOFTWARE REQUIREMENTS A. FUNCTIONAL REQUIREMENTS B. NON-FUNCTIONAL REQUIREMENTS C.	9-10
4.	SYSTEM STUDY A. FEASIBILITY STUDY	11-12
5.	SOFTWARE REQUIREMENT ANALYSICS A. DEFINING THE PROBLEM B. PHASES OF APPLICATION C. MODULES AND THEIR FUNCTIONALITIES D.	13-16
6.	SOFTWARE ENGINEERING PARADIGN	17-23
7.	PERT CHART &GAND CHART	24
8.	SOFTWARE DESIGN A. UML DIAGRAMS B. DATABASE DESIGN C. DATA FLOW DIAGRAM D.	25-39
9.	SOFTWARE ENVIRONMENT	40-49
10.	IMPLEMENTATIONS	50-51
11.	TESTING	52-54
12.	TEST CASE ANALYSIS	55-56
13.	CONCLUSION	57
14.	FURTHER ENHANCEMENTS	58
15.	APPENDIX A:BIBILOGRAPHY	59
16.	APPENDIX B:SAMPLE SOURCE CODE	60-80
17.	APPENDIX C:OUTPUT SCREEN	81-88

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INTRODUCTION

1. PROJECT DESCRIPTION

The continuous growth in the size and use of the World Wide Web imposes new methods of design and development of online information services. Most Web structures are becoming complicated and users often miss the goal of their inquiry, or receive ambiguous results when they try to navigate through them which leads a user to untrusted websites, products and links. On the other hand, the E-business sector is rapidly evolving and the needs for web market places that anticipate the needs of the customers and the trust towards a product are equally more evident than ever. While people are enjoying the benefits from online trading, criminals are also taking advantages to conduct fraudulent activities against honest parties to obtain illegal profits. Therefore, the requirement for predicting user needs and trust providence towards a product in order to improve the usability and user retention of a website can be addressed by personalizing and using a fraud product detection system.

The application for storage of data has been planned to use the MySQL and all the user interfaces has been designed using the JSP Technologies. The application takes care of different modules and their associated functionalities as per the applicable strategies.

2. FRAUD PRODUCT DETECTION

Where it was once acceptable for companies to sell their products to very defined and localized markets within certain logical timeframes, the advent of online shopping has completely redefined the way companies now market themselves in order to establish a market presence. However, the introduction of this dynamic medium of conducting business has brought with it its own complex set of problems. Although many businesses are well placed to be able to capture the emerging markets that electronic commerce can open up, factors such as widespread concerns about fraud and Internet security have greatly hindered online business prospects. It must be noted that these concerns are shared by both consumers as well as corporate organizations, which stand to lose sizable amounts from fraudulent activities. Fraud product detection allows a user or a customer to know about the product trustworthiness through the other user's feedback for that product.

3. WEB PERSONALISATION

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Web personalization is defined as any action that adapts the information or services provided by a website to the needs of a user or a set of users, taking advantage of the knowledge gained from other users' behaviour and individual interests in combination with the content or it can also be defined as a process of gathering and storing information, analysing the information, and taking the decision based on the analysis.

Fraud detection and web personalization are the key technologies needed in various e-business applications to,

a. Manage customer organization relationships
b. Promote products
c. Manage Web site content
d. Provide knowledge to the user about the product.

4. PROJECT PURPOSE

a. Improves Customer Seller relationship in our application, more productive and engaging.
b. Valuable to you and your organization, because it drives desired business results such as increasing visitor response or promoting customer retention.
c. Most importantly, keep the process simple. Stay focused on the business goals, tackle manageable projects, measure the success or failure of your changes, and learn from your mistakes.
d. Improves the productivity by simplifying access to information
e. More likely to increase sales of trusty companies

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ABSTRACT

We consider the problem of building online machine-learned models for detecting auction frauds in e-commerce web sites. Since the emergence of the World Wide Web, online shopping and online auction have gained more and more popularity. While people are enjoying the benefits from online trading, criminals are also taking advantages to conduct fraudulent activities against honest parties to obtain illegal profit. Hence proactive fraud-detection moderation systems are commonly applied in practice to detect and prevent such illegal and fraud activities. Machine-learned models, especially those that are learned online, can catch frauds more efficiently and quickly than human-tuned rule-based systems. In this paper, we propose an online probit model framework which takes online feature selection, coefficient bounds from human knowledge and multiple instances learning into account simultaneously. By empirical experiments on a real-world online auction fraud detection data we show that this model can potentially detect more frauds and significantly reduce customer complaints compared to several baseline models and the human-tuned rule-based system.

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HARDWARE AND SOFTWARE REQUIREMENTS

1. HARDWARE REQUIREMENTS

Processor	Pentium
Ram	256MB

2. SOFTWARE REQUIREMENTS

Web server	Apache Tomcat Server
Operating system	Windows
Language	JSP (Java Server Pages)
database	MySQL

One of the fundamental objectives of any project is to collect both the functional and non-functional requirements. These need to be kept in balance and harmony, as the project progresses.

a. Functional Requirements

These are the statements of services that the system should provide, how the system should react to inputs and how the system should behave situations.

b. Functional requirements

These requirements specify criteria that can be used to judge the operation of a system, rather than specific behaviours. These requirements are often called qualities of a system. Some of the non-functional requirements include performance, security, user-interface etc. Below is the chart of requirements which include both functional and non-functional

1. Name	Proactive Moderation and A personalized System for Fraud Product Detection

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2. purpose	To make user available time with trust worthy products without spending much of the time in knowing about the product
3. inputs	Ratings, Feedback
4. outputs	Trustworthy products are made available
5. security	Usernames and password to each user
6. user interface	Buttons and links on the screen allow the user to control the system.

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SYSTEM STUDY

1. FEASIBILITY STUDY

The feasibility of the project is analysed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

a. ECONOMICAL FEASIBILITY
b. TECHNICAL FEASIBILITY
c. SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only

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minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

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SOFTWARE REUQIREMENT ANALYSICS

1. DEFINING THE PROBLEM

a. Existing problem

The traditional online shopping business model allows sellers to sell a product or service at a preset price, where buyers can choose to purchase without any information related to the quality of the product. This makes user to make extra time in knowing the information about the product based on his/her interests which may also frustrate the user and sometimes lead the user in not buying which indirectly reduces the sales of website.

b. Proposed system

The proposed system delivers the right content to the right person to maximize immediate and future business opportunities. This also increases the productivity and sales by simplifying access to information there by reducing the time to decide whether to trust the product or not.

2. PHASES OF APPLICATION

This application requires implicitly or explicitly collecting visitor purchase information and leveraging that knowledge in your content delivery framework to manipulate what information you present to users.

The steps include:

- Collection of data
- Analysis of the collected data
- Determination of the actions that should be performed.

a. Collection of data

Whatever method is eventually used to process the data, information about user's behaviour and products must first be collected.

Explicit data collection refers to any method where the user is asked to provide feedback or information about product. Often, this begins after a user purchases a product or used a product. The feedback includes the rating for good, poor delivery, poor manufacturing or usage or general text about the product. All the information will be collected from different users and the status of the product will be updates whether to trust or not.

b. Analysis of the collected data

The ways that are employed in order to analyze the collected data include are

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- Rule-based features:

Human experts with years of experience created many rules to detect whether a user is fraud or not. It checks whether the product has been or complained as untrusting or fraud.

The trust for particular product(X) can be calculated (in %) by

Trust(X) =100-Fraud(X)

Fraud(X) =No of complaints(X)/ (No of users(X)*0.01)

- Selective labelling

If the fraud score is above a certain level, the case will enter a queue for further investigation by human experts and the cases whose fraud score are below are determined as clean by the human expert.

c. Decision making/Final Recommendation

The decision or the final recommendation after analysis part is to decide whether to ban the product or to trust the product. If the product is banded by the admin then no user can view or buy the product hence providing the user only the trustworthy products.

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MODULES AND THEIR FUNCTIONALITIES

The system has been classified into the following modules after a careful analysis,

1. Customer Module
2. Seller Module
3. Administrative Module
4. Complaint Filing
5. Fraud churn

CUSTOMER MODULE

A customer is one of the users who wish to shop online. For this purpose the customer will be provided with a personal account through registration. After successful registration, he will be provided with a gallery of different products from different sellers which include the product name, price, sellers' name etc. While buying a product a customer can view the percent of trustworthiness towards the product given by other users. After purchasing, a customer can also file complaint on that product where he feels uncomfortable provided with some options like

1. Products purchased by the buyer are not delivered by the seller.
2. The delivered products do not match the descriptions that were posted by sellers.
3. Malicious sellers may even post non-existing items with false description to deceive buyers
4. General feedback as a complaint

SELLER MODULE

The seller module includes different sellers who wish to sell their products. The seller needs to be approved by administrator after a seller submits his registration. A seller can add or delete or modify information about different items.

The different functionalities for seller are

1. Can add a new a product
2. Can delete a product
3. Can place new offers to the product
4. Can modify information related to the product such as price, basic information etc...

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ADMIN MODULE

The administrative module includes an admin who acts as an intermediator between seller and the customer. An Admin is responsible to maintain the website information giving a trust to the customers. When a complaint is filed in the customer module, the admin takes the final decision whether to ban the product. If the admin feels all the products from seller mostly are not trusted he can also remove the seller and his related products.

COMPLAINT FILLING

Buyers can file complaints to claim loss if they are recently deceived by fraudulent sellers. The Administrator views the various types of complaints and the percentage of various type complaints. The complaints values of a products increase some threshold value the administrator set the trust ability of the product as Untrusted or banned. If the products set as banned, the user cannot view the products in the website.

FRAUD CHURN

In this module admin takes the decision whether to continue the seller to sell the products or not. When some products are labelled as fraud by human experts, it is very likely that the seller is not trustable and the products too. Hence all the items submitted by the same seller are labelled as fraud too. So, the fraudulent seller along with his/her cases will be removed from the website immediately once detected.

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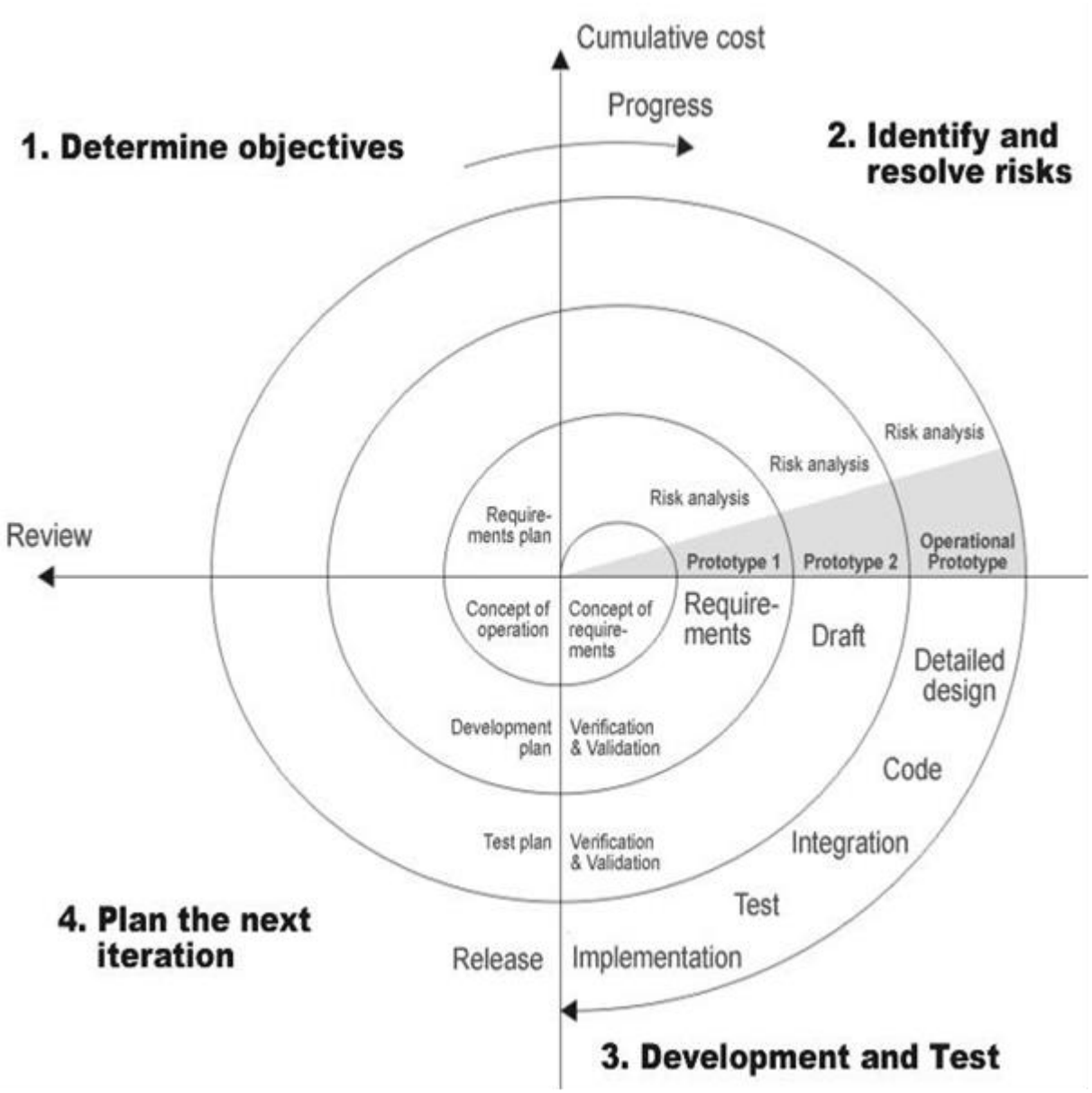
SOFTWARE ENGINEERING PARADIGM

PROCESS MODEL USED WITH JUSTIFICATION

SDLC (Spiral Model):



SDLC is nothing but Software Development Life Cycle. It is a standard which is used by software industry to develop good software.

STAGES IN SDLC:

- ❖ Requirement Gathering
- ❖ Analysis
- ❖ Designing
- ❖ Coding
- ❖ Testing
- ❖ Maintenance

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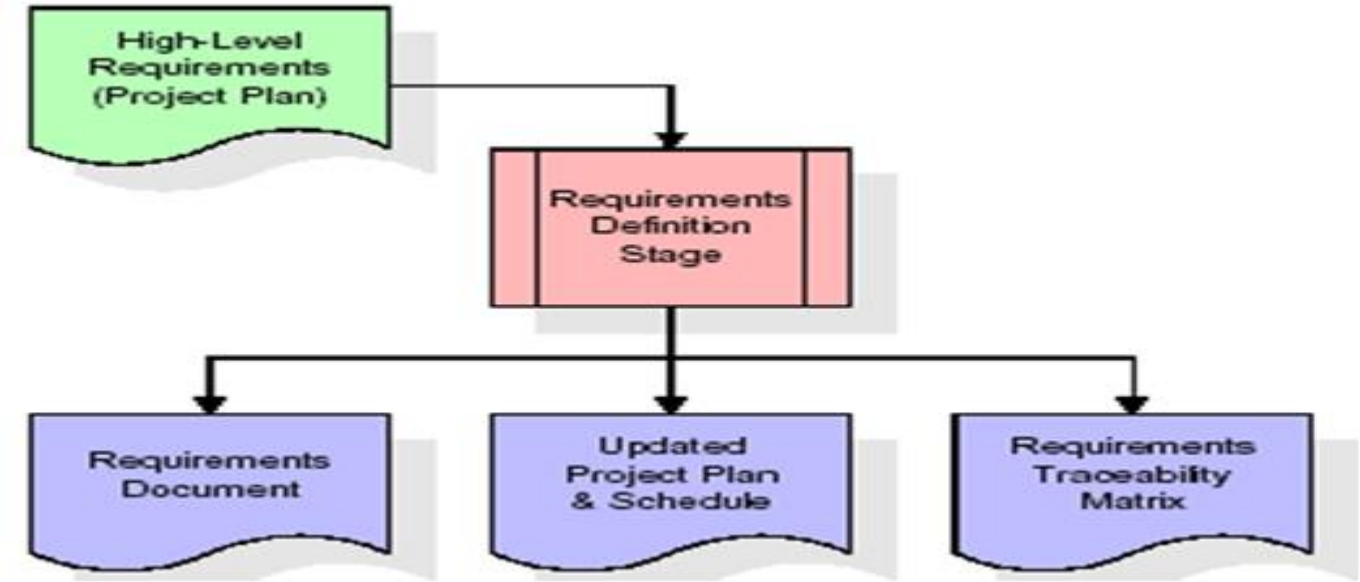
REQUIREMENT GATHERING

The requirements gathering process takes as its input the goals identified in the high-level requirements section of the project plan. Each goal will be refined into a set of one or more requirements. These requirements define the major functions of the intended application, define operational data areas and reference data areas, and define the initial data entities. Major functions include critical processes to be managed, as well as mission critical inputs, outputs and reports. A user class hierarchy is developed and associated with these major functions, data areas, and data entities. Each of these definitions is termed a Requirement. Requirements are identified by unique requirement identifiers and, at minimum, contain a requirement title and textual description.



These requirements are fully described in the primary deliverables for this stage: the Requirements Document and the Requirements Traceability Matrix (RTM). The requirements document contains complete descriptions of each requirement, including diagrams and references to external documents as necessary. Note that detailed listings of database tables and fields are not included in the requirements document.

The title of each requirement is also placed into the first version of the RTM, along with the title of each goal from the project plan. The purpose of the RTM is to show that the product components developed during each stage of the software development lifecycle are formally connected to the components developed in prior stages.

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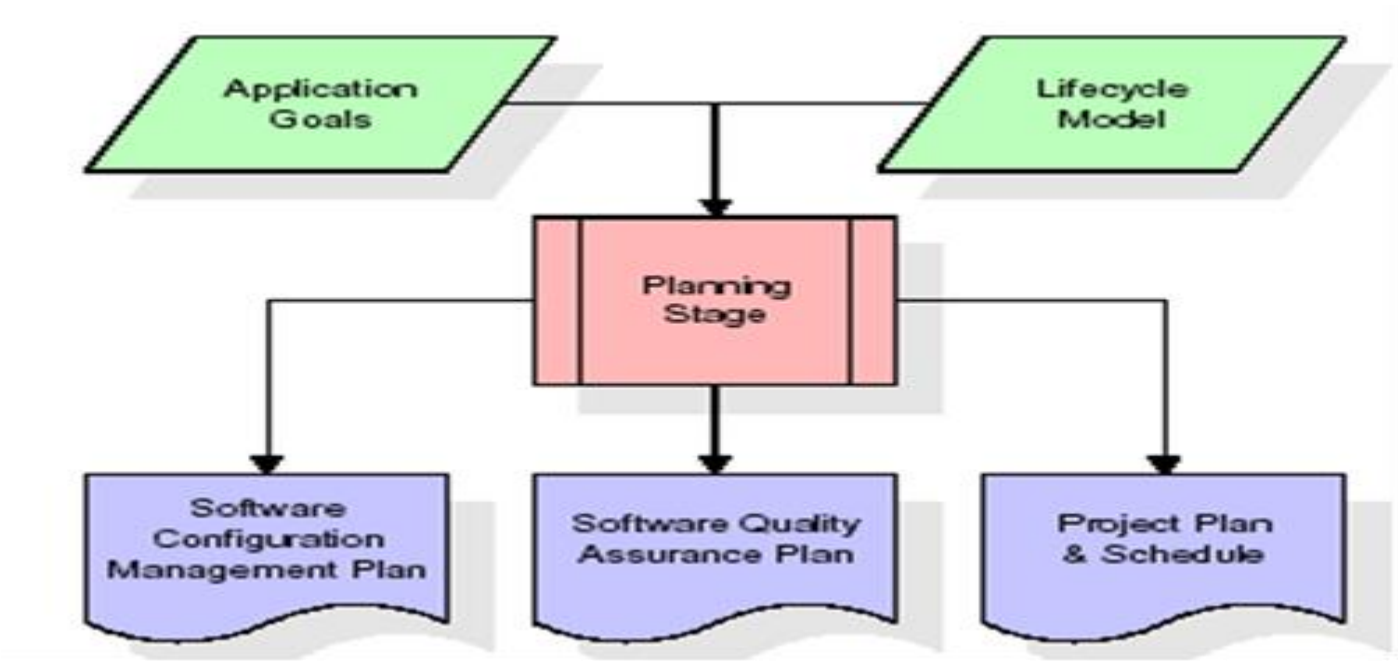
In the requirements stage, the RTM consists of a list of high-level requirements, or goals, by title, with a listing of associated requirements for each goal, listed by requirement title. In this hierarchical listing, the RTM shows that each requirement developed during this stage is formally linked to a specific product goal. In this format, each requirement can be traced to a specific product goal, hence the term requirements traceability.

The outputs of the requirements definition stage include the requirements document, the RTM, and an updated project plan.

- Feasibility study is all about identification of problems in a project.
- No. of staff required to handle a project is represented as Team Formation, in this case only modules are individual tasks will be assigned to employees who are working for that project.
- Project Specifications are all about representing of various possible inputs submitting to the server and corresponding outputs along with reports maintained by administrator

ANALYSIS

The planning stage establishes a bird's eye view of the intended software product, and uses this to establish the basic project structure, evaluate feasibility and risks associated with the project, and describe appropriate management and technical approaches.



The most critical section of the project plan is a listing of high-level product requirements, also referred to as goals. All the software product requirements to be developed during the requirements definition stage flow from one or more of

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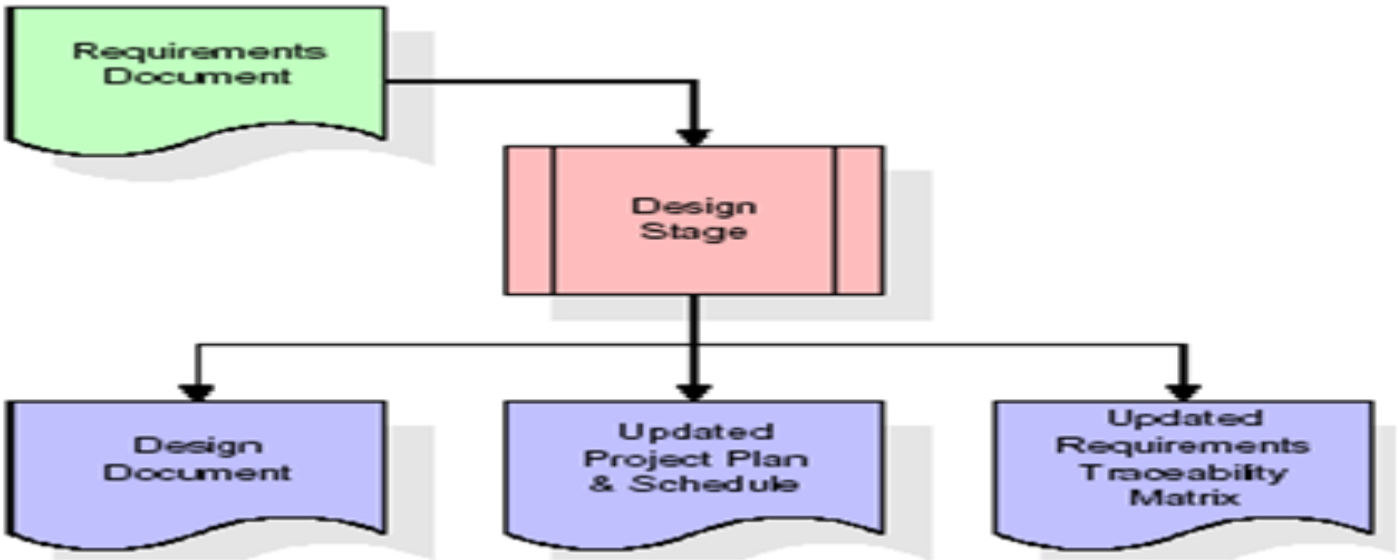
these goals. The minimum information for each goal consists of a title and textual description, although additional information and references to external documents may be included. The outputs of the project planning stage are the configuration management plan, the quality assurance plan, and the project plan and schedule, with a detailed listing of scheduled activities for the upcoming Requirements stage, and high-level estimates of effort for the out stages.

DESIGNING

The design stage takes as its initial input the requirements identified in the approved requirements document. For each requirement, a set of one or more design elements will be produced as a result of interviews, workshops, and/or prototype efforts. Design elements describe the desired software features in detail, and generally include functional hierarchy diagrams, screen layout diagrams, tables of business rules, business process diagrams, pseudo code, and a complete entity-relationship diagram with a full data dictionary. These design elements are intended to describe the software in sufficient detail that skilled programmers may develop the software with minimal additional input.



When the design document is finalized and accepted, the RTM is updated to show that each design element is formally associated with a specific requirement. The outputs of the design stage are the design document, an updated RTM, and an updated project plan.

CODING

The development stage takes as its primary input the design elements described in the approved design document. For each design element, a set of one or more software artifacts will be produced. Software artifacts include but are not limited to menus, dialogs, data management forms, data reporting formats, and

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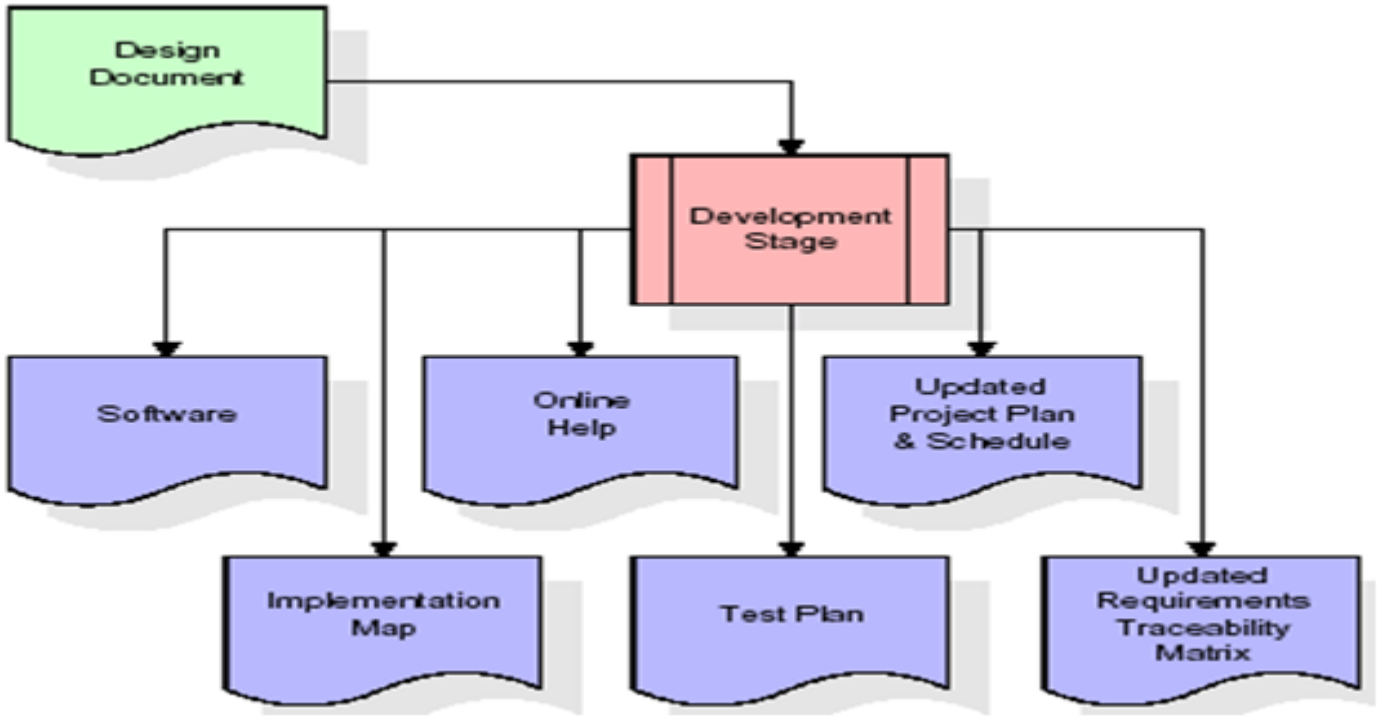
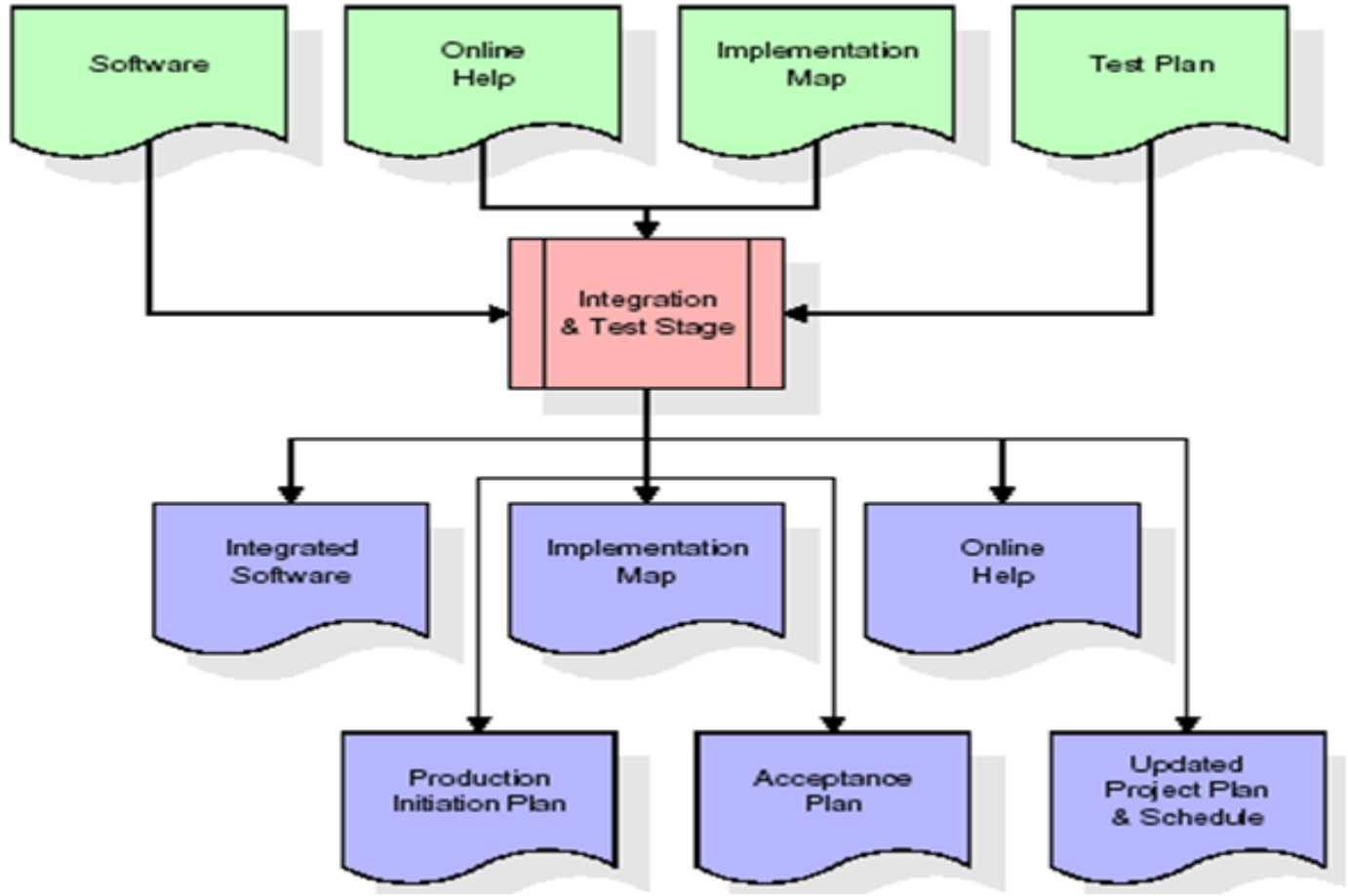
specialized procedures and functions. Appropriate test cases will be developed for each set of functionally related software artifacts, and an online help system will be developed to guide users in their interactions with the software.



The RTM will be updated to show that each developed artifact is linked to a specific design element, and that each developed artifact has one or more corresponding test case items. At this point, the RTM is in its final configuration. The outputs of the development stage include a fully functional set of software that satisfies the requirements and design elements previously documented, an online help system that describes the operation of the software, an implementation map that identifies the primary code entry points for all major system functions, a test plan that describes the test cases to be used to validate the correctness and completeness of the software, an updated RTM, and an updated project plan.

TESTING

During the integration and test stage, the software artifacts, online help, and test data are migrated from the development environment to a separate test environment. At this point, all test cases are run to verify the correctness and completeness of the software. Successful execution of the test suite confirms a robust and complete migration capability. During this stage, reference data is finalized for production use and production users are identified and linked to their appropriate roles. The final reference data (or links to reference data source files) and production user list are compiled into the Production Initiation Plan.



The outputs of the integration and test stage include an integrated set of software, an online help system, an implementation map, a production initiation plan that describes reference data and production users, an acceptance plan which contains the final suite of test cases, and an updated project plan.

INSTALLATION

During the installation and acceptance stage, the software artifacts, online help, and initial production data are loaded onto the production server. At this point, all test cases are run to verify the correctness and completeness of the software. Successful execution of the test suite is a prerequisite to acceptance of the software by the customer.

After customer personnel have verified that the initial production data load is correct and the test suite has been executed with satisfactory results, the customer formally accepts the delivery of the software.

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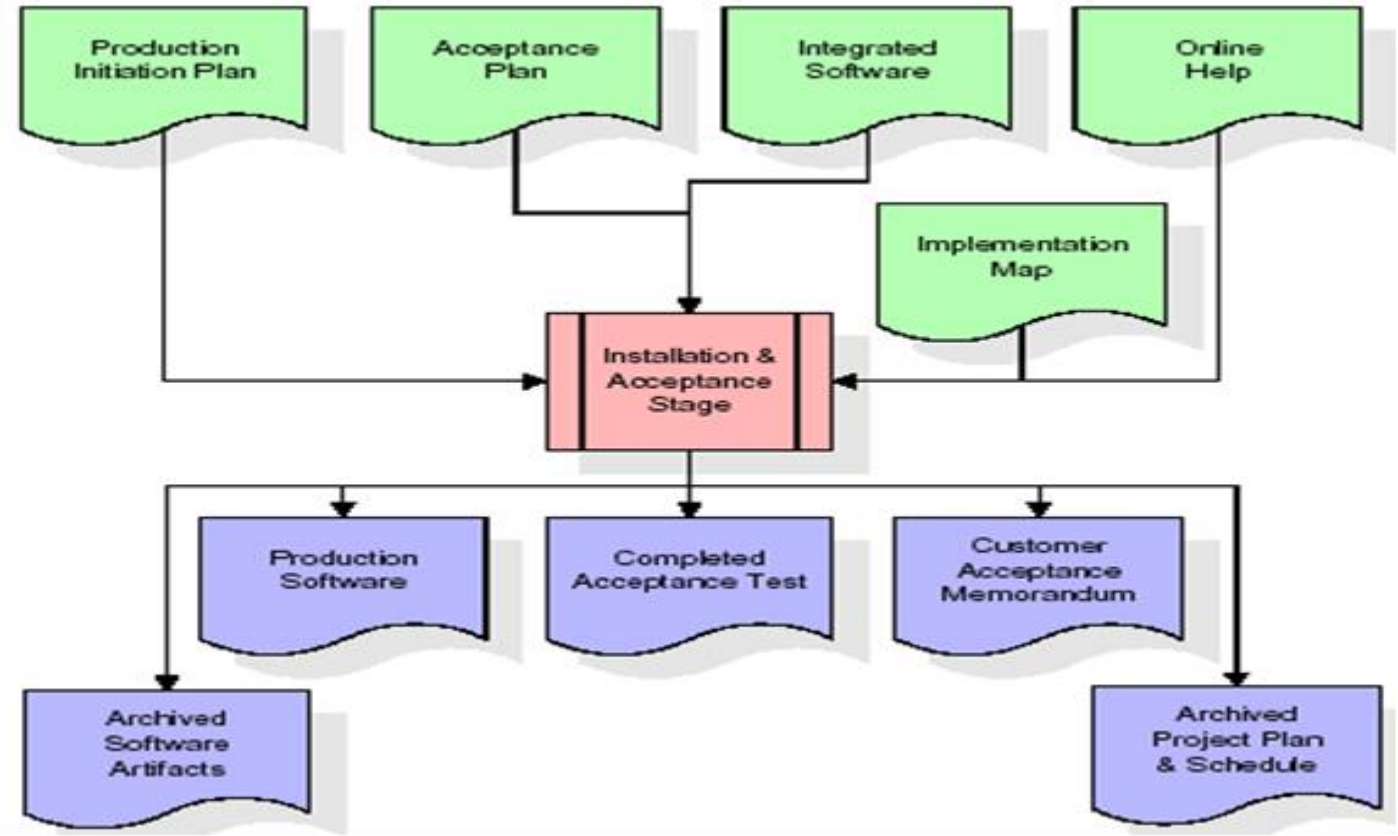

The primary outputs of the installation and acceptance stage include a production application, a completed acceptance test suite, and a memorandum of customer acceptance of the software. Finally, the PDR enters the last of the actual labor data into the project schedule and locks the project as a permanent project record. At this point the PDR "locks" the project by archiving all software items, the implementation map, the source code, and the documentation for future reference.

MAINTAINANCE

Outer rectangle represents maintenance of a project, Maintenance team will start with requirement study, understanding of documentation later employees will be assigned work and they will undergo training on that particular assigned category.

For this life cycle there is no end, it will be continued so on like an umbrella (no ending point to umbrella sticks).

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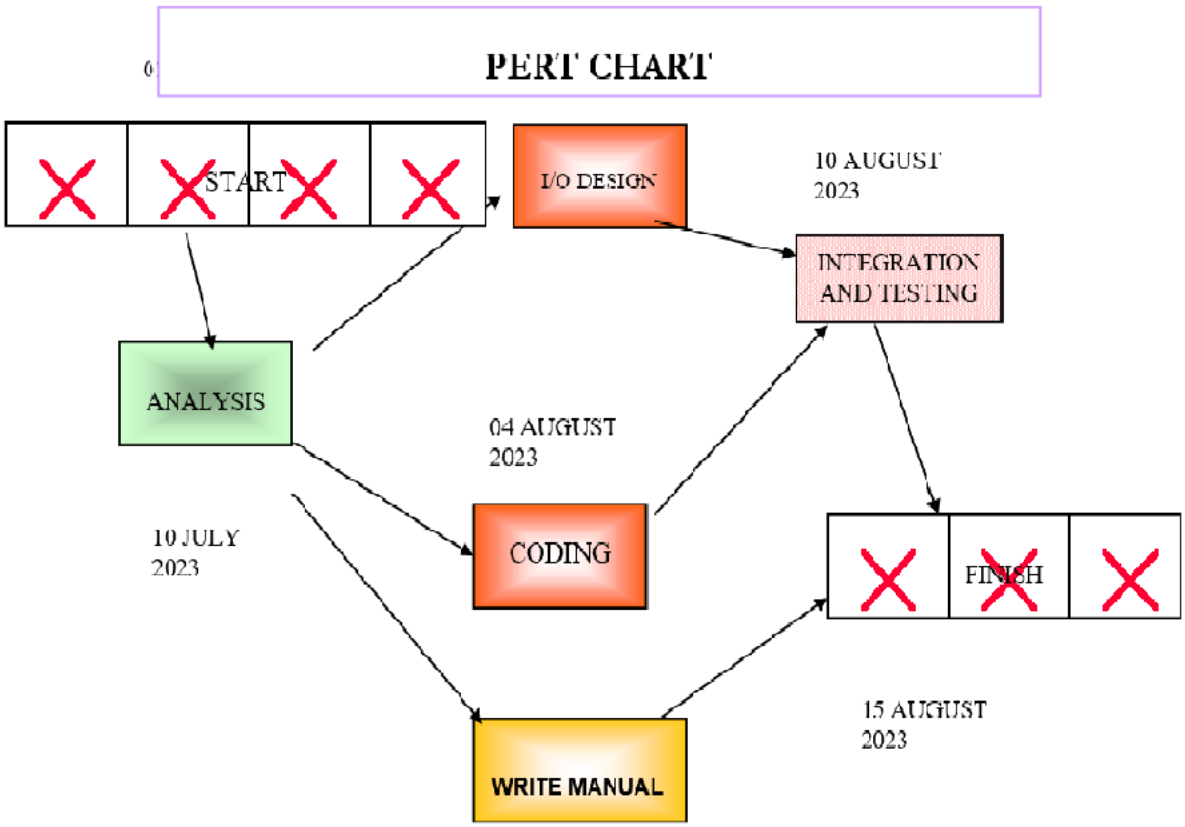
PERT CHART&GAND CHART

PERT CHART

Program Evaluation Review Technique, PERT can be both a cost and a time management system. PERT is organized by events and activities or tasks. PERT has several advantages over bar charts and is likely to be used with more complex projects. One advantage of PERT is that it is scheduling device that also shows graphically which tasks must be completed before others are begun.

Also, by displaying the various task paths, PERT enables the calculation of a critical path. Each path consists of combinations of tasks, which must be completed. PERT controls time and cost during the project and also facilitates finding the right balance between completing a project on time and completing it within the budget.



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SOFTWARE DESIGN

1. UML DESIGN

Unified Modelling Language (UML) is a standardized general-purpose modeling language in the field of object-oriented software engineering. The Unified Modelling Language includes a set of graphic notation techniques to create visual models of object-oriented software-intensive systems.

Unified Modelling Language is used to specify, visualize, modify, construct and document the artifacts of an object-oriented software-intensive system under development.

We have used three types of diagrams to describe the modules in our project. They are

- Use case diagrams
- Sequence diagrams
- Class diagrams

a. USE CASE DIAGRAMS

Use case diagrams model the functionality of system using actors and use cases. These diagrams are central to modelling the behaviour of a system, a subsystem, or a class.

b. SEQUENCE DIAGRAMS

A sequence diagram is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence chart. Sequence diagram are sometimes called Event diagrams, event scenarios and timing diagrams.

c. CLASS DIAGRAMS

Class Diagrams is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (methods) and the relationships among the classes. It can also be described as a set of objects that share the same attributes, operations, relationships, and semantics.

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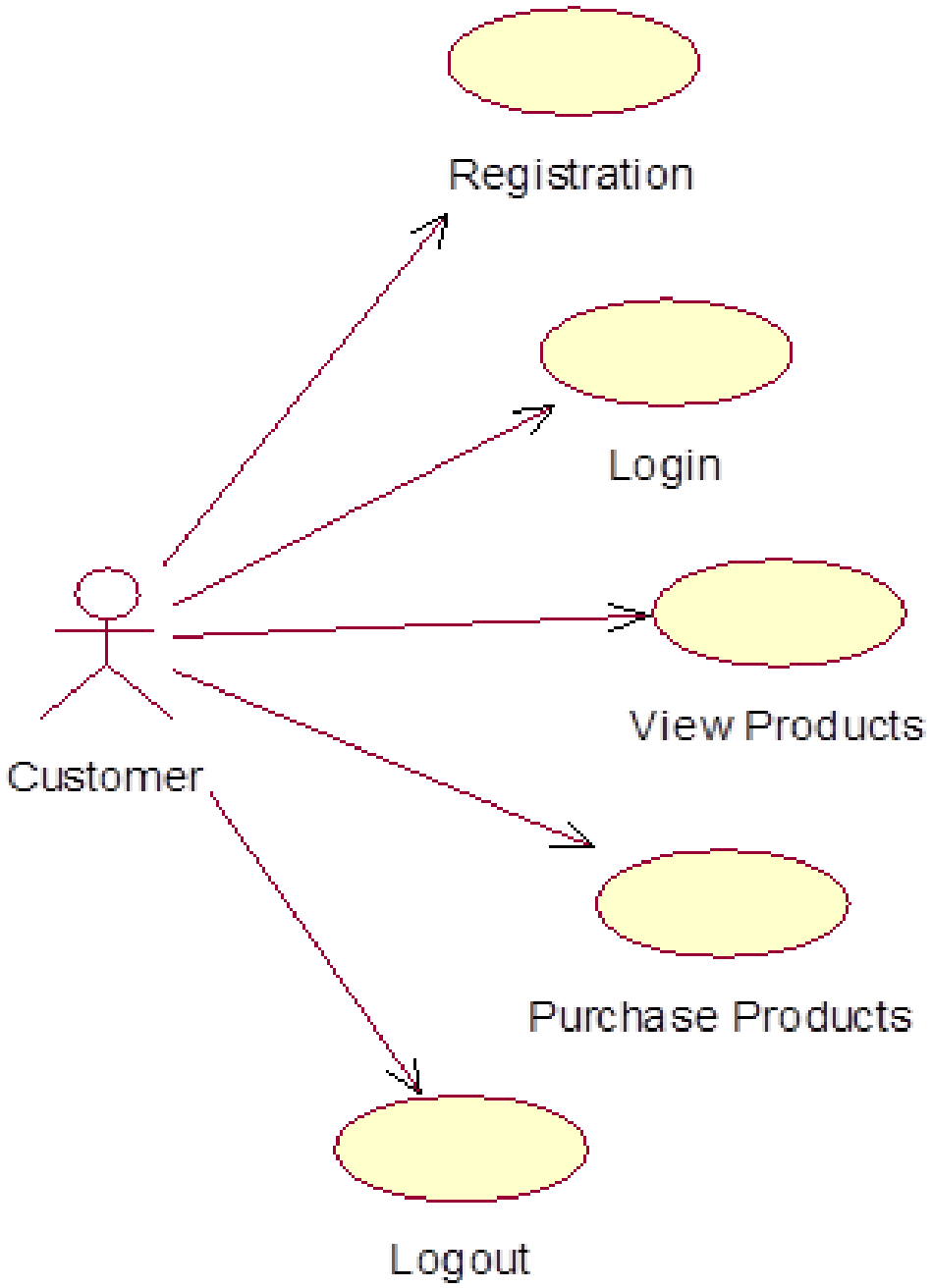
USE CASE DIAGRAM FOR CUSTOMER PURCHASE



A customer is provided with a personal account through registration process. Once the account has been created, he can login. The customer will be provided with a gallery of products in which he can select and purchase the products.

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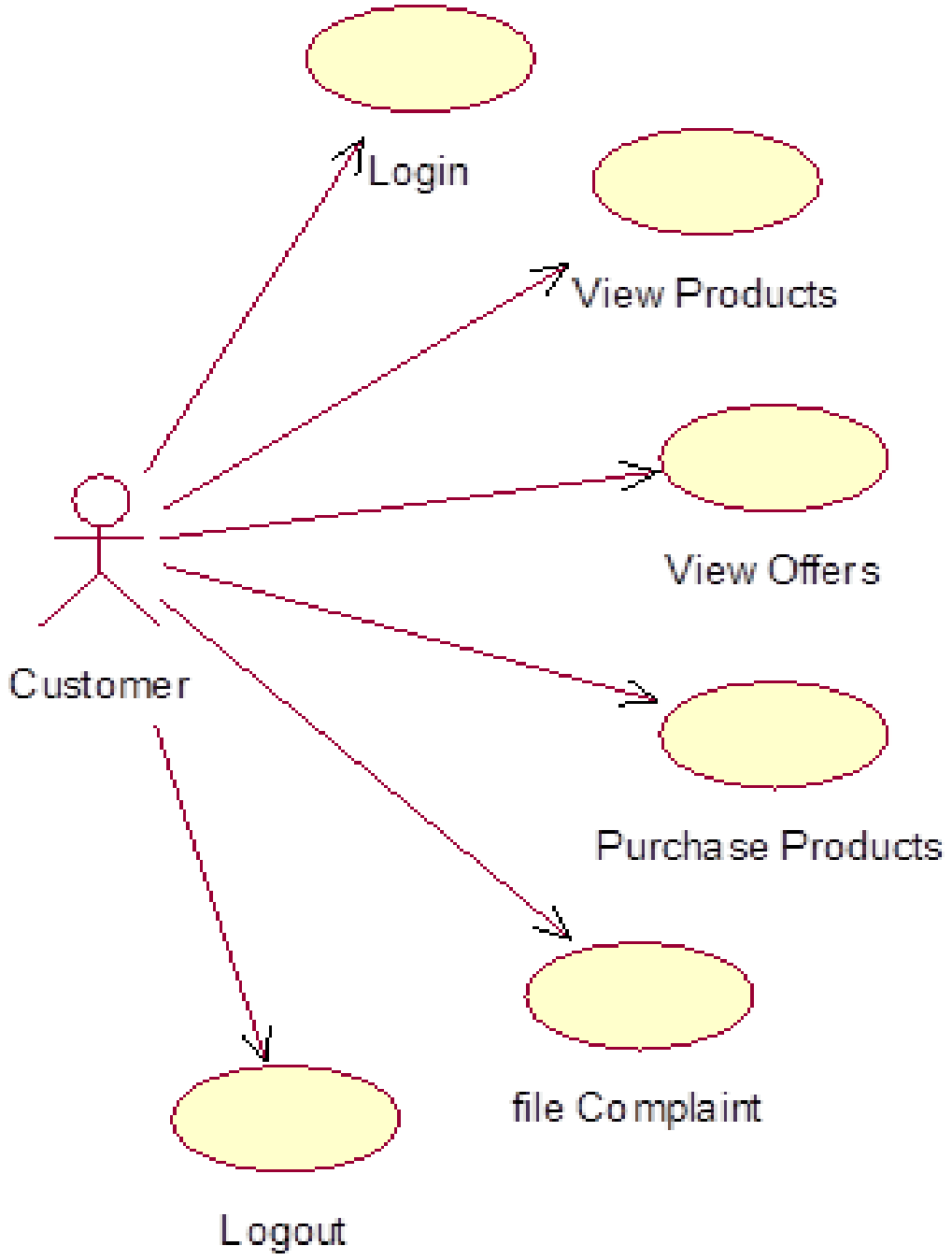
USECASE DIAGRAM FOR CUSTOMER COMPLAINT



A customer is provided with a personal account through registration process. Once the account has been created, he can login. The customer will be provided with a gallery of products in which he can select and purchase the products. After purchase the customer can file a complaint the product in any aspect.

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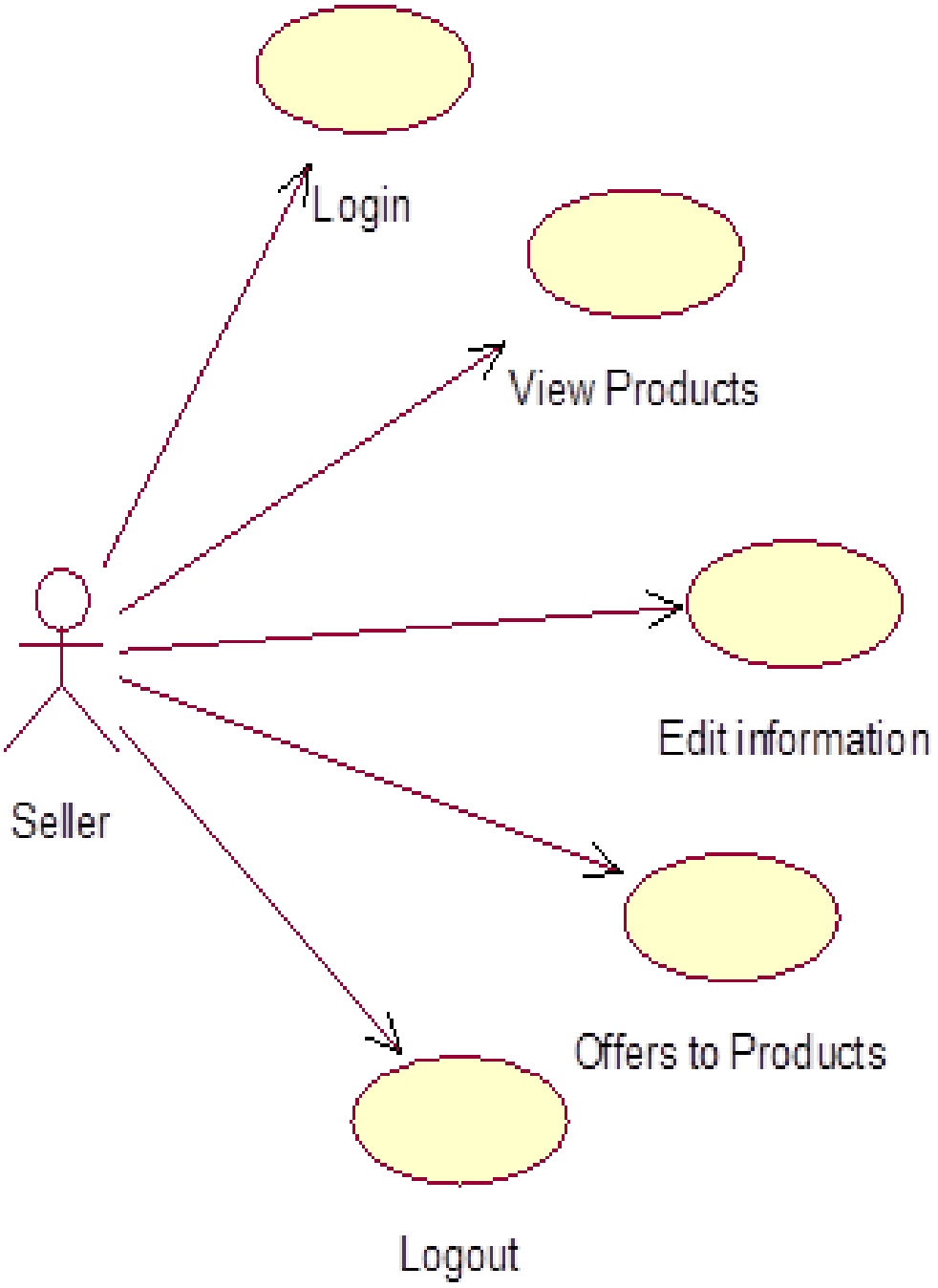
USECASE DIAGRAM FOR SELLER



A Seller can add or delete or modify information about different items based on the category. A seller can also provide special offers to the customers to increase the sales.

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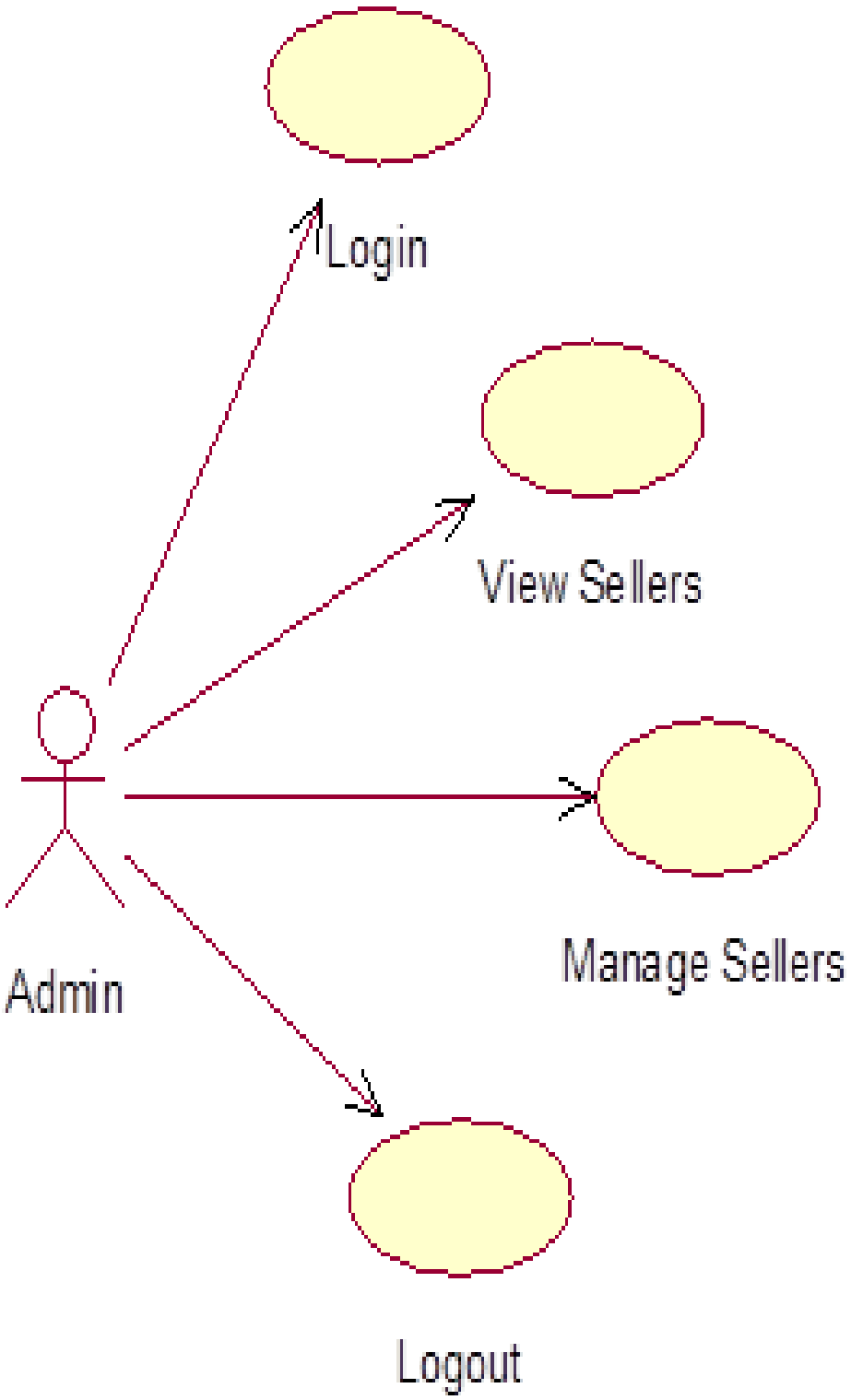
USECASE DIAGRAM FOR ADMIN TO MANAGE SELLERS



The administrator maintains the website activities by modifying/adding or deleting the sellers based on the products they sell.

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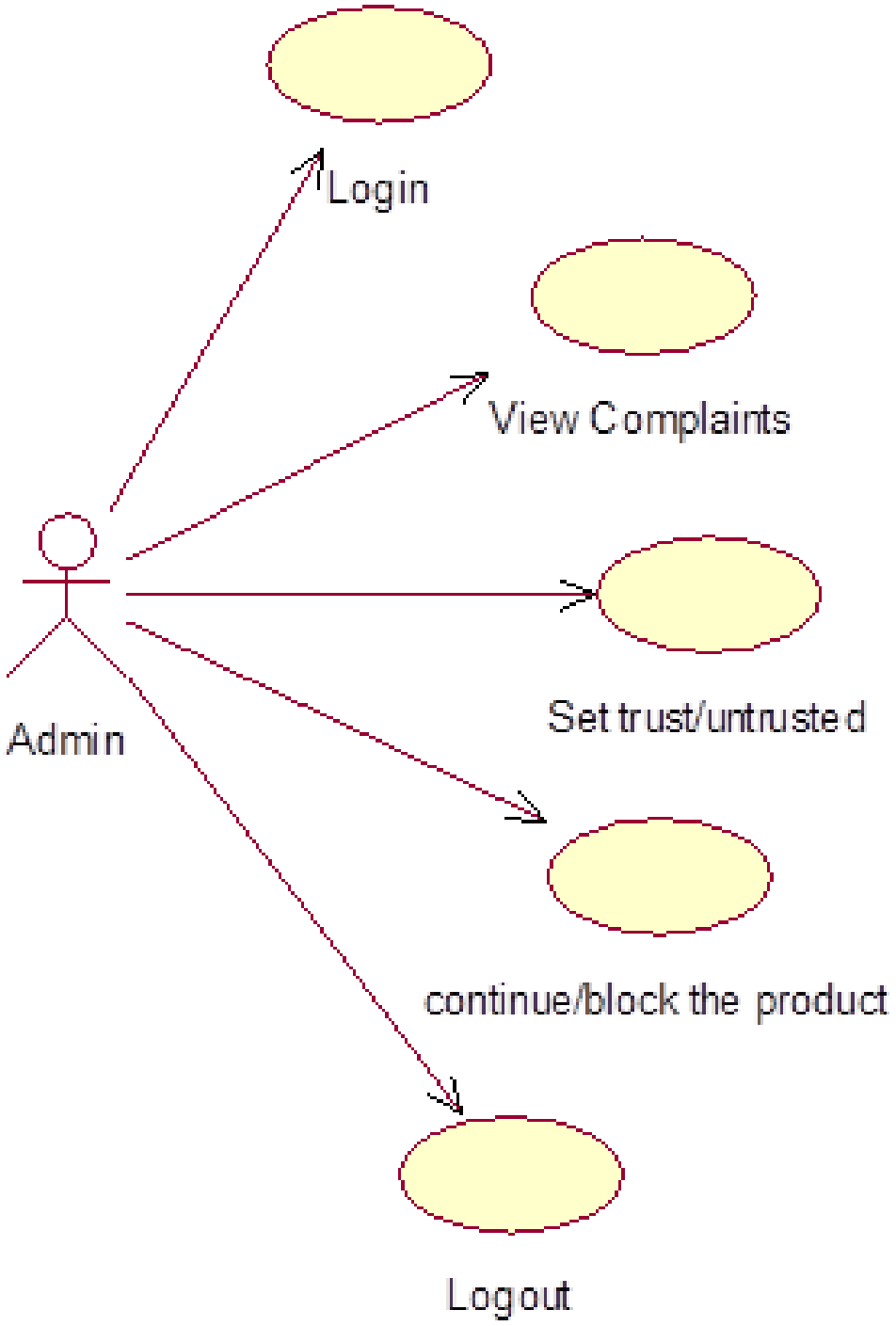
USECASE DIAGRAM FOR ADMIN



When a complaint is filed in the customer module the admin takes the final decision whether to ban the product or trust or to give sometime. If the admin feels all the products from seller mostly are not trusted he can also remove the seller and his related products.

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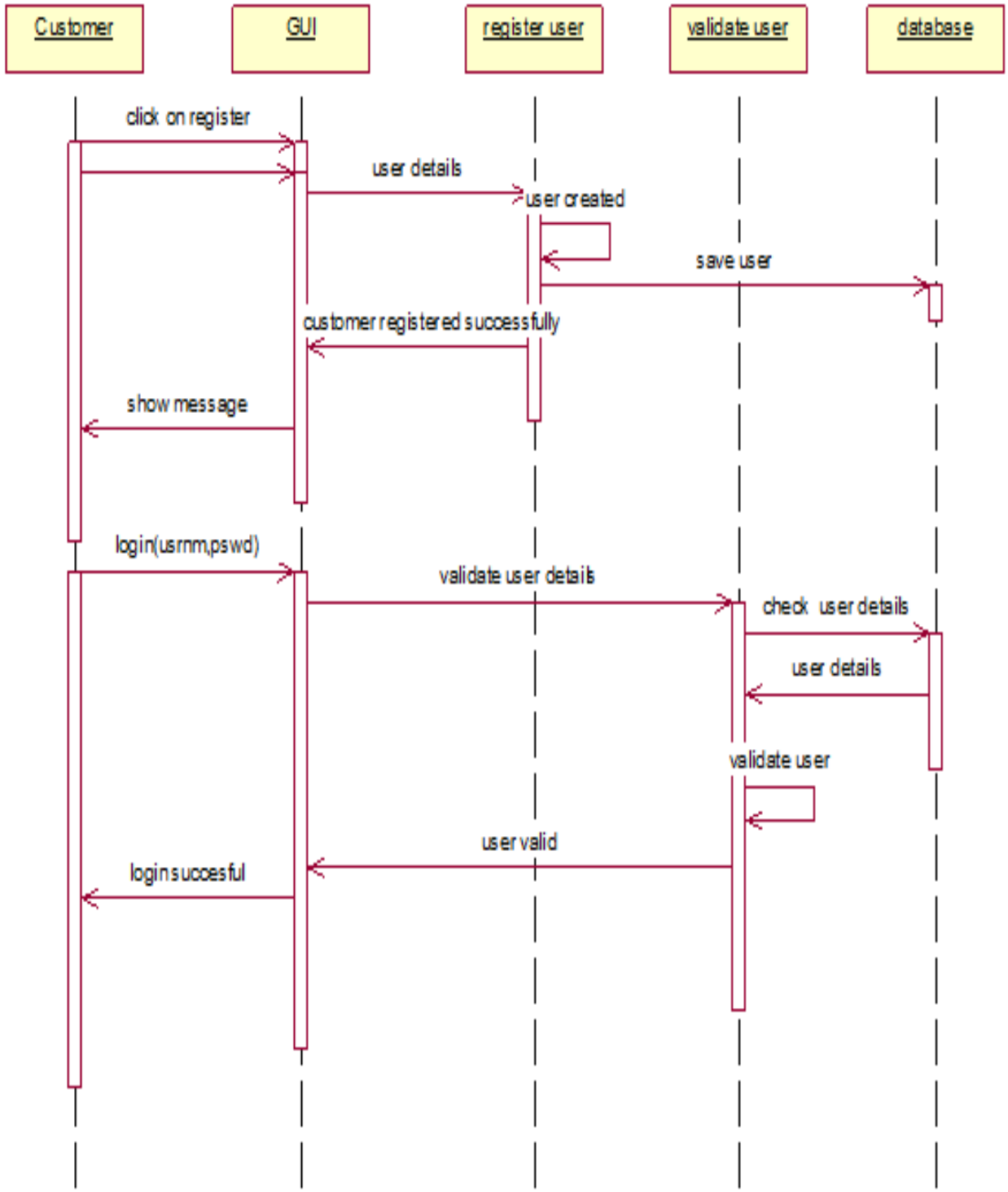
SEQUENCE DIAGRAM FOR CUSTOMER REGISTRATION/LOGIN



For registration, the details must be stored properly and then account will be created for a user. While logging in, a customer details needs to be validated with the previous data which has been stored during registration.

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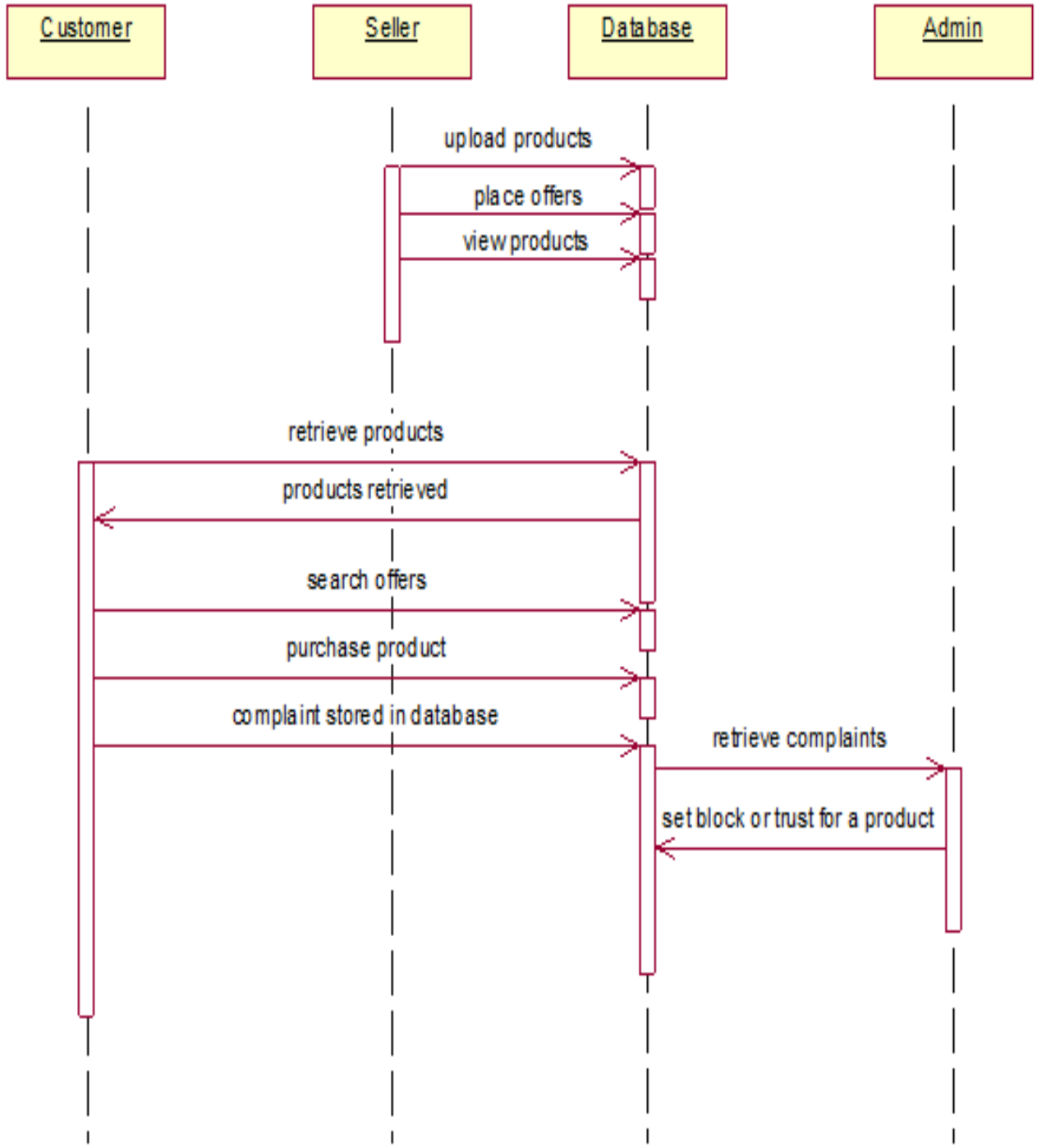
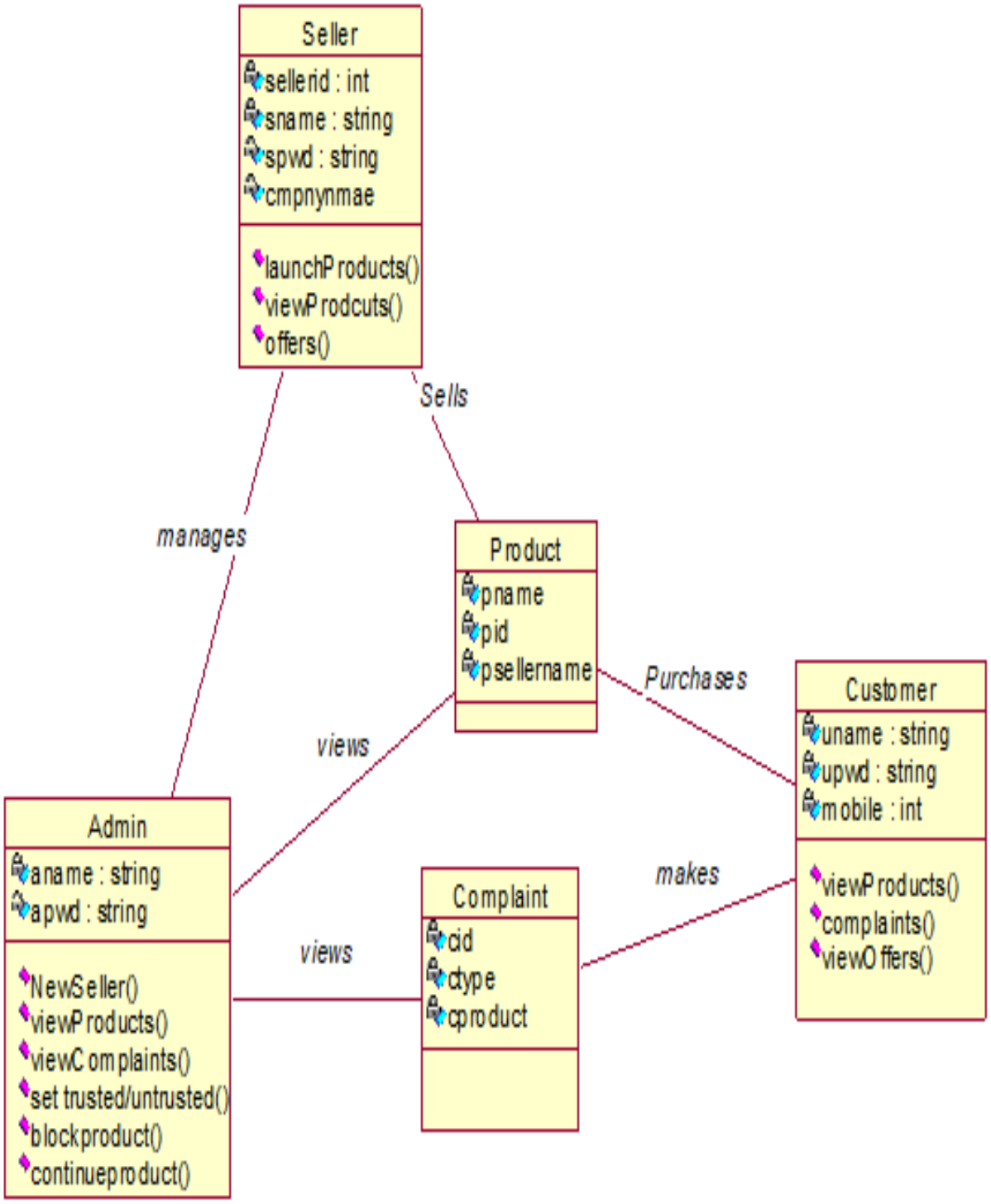
SEQUENCE DIAGRAM FOR APPLICATION



A customer views the offers and products and on interest buys the products. The seller can update/add/delete the product and provides offers to customers. The admin manages the seller and takes the decision of which provided need to be in the website.

`

CLASS DIAGRAM FOR APPLICATION



The Class diagram shows different classes and how they are related. The seller who sells the product will be managed by the admin who views the products and complaints filed by the customer.

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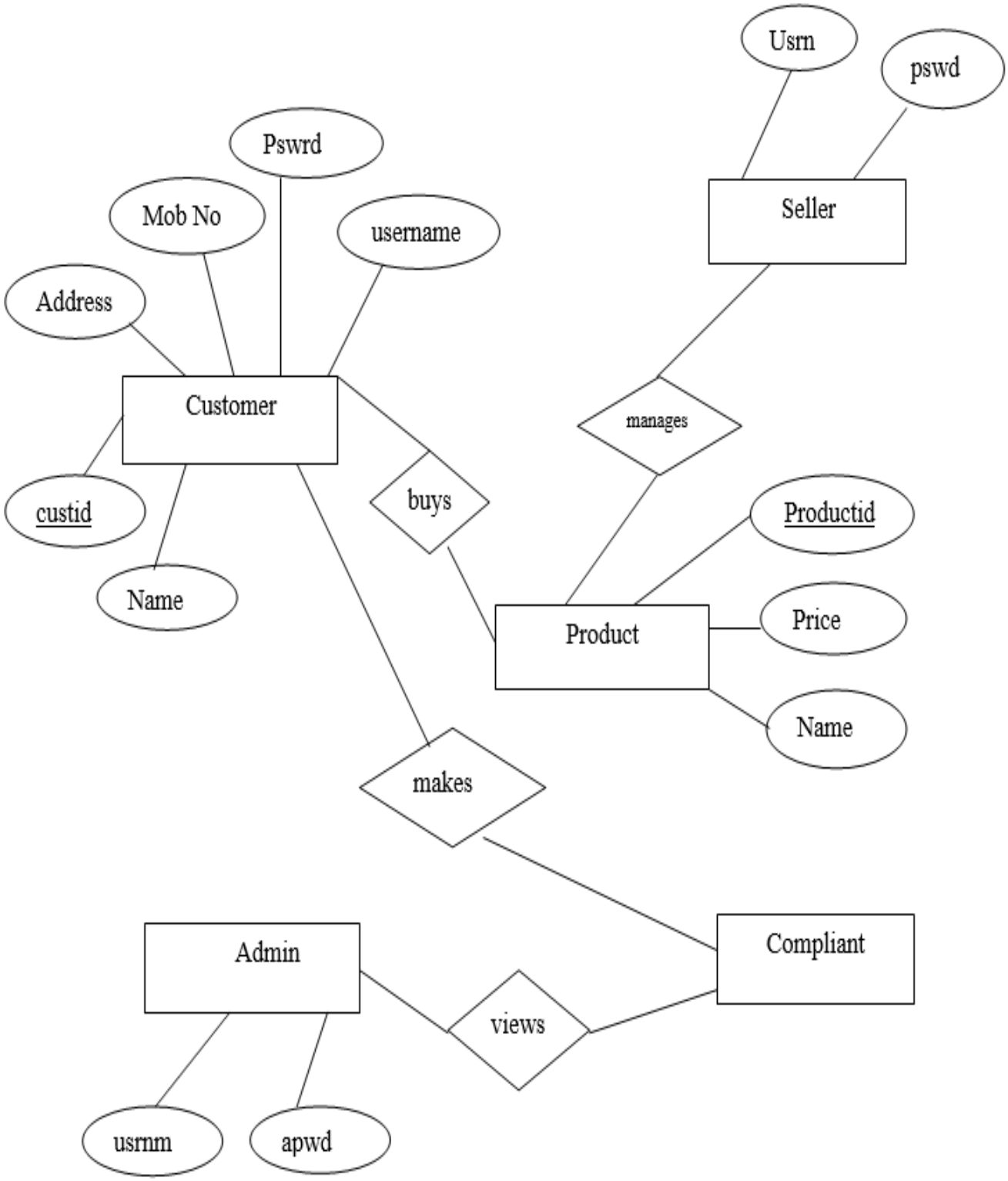
E-R DIAGRAM



The diagram shows how different entities are related. N number of customers can buy N products. One admin manages N products who also maintain Nsellers and N sellers can sell N product which can be purchased by N customers. A customer can also file complaint but only 1 complaint to one product. This is the how the entire application works

`

2. DATABASE DESIGN

Database design is the process of producing a detailed data model of a database. This logical data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design in a Data Definition Language, which can then be used to create a database. A fully attributed data model contains detailed attributes for each entity.

ADMIN TABLE

ATTRIBUTES	TYPE
USERID	VARCHAR
PASS	VARCHAR

The above table consists of admin login details. These values will be further used in validating an admin detail avoiding the unauthorized people using the account

OFFERS TABLE

ATTRIBUTES	TYPE
PID	Numeric
COMMNAME	Varchar
PRONAME	Varchar
WARDATE	Varchar
PRORATE	Varchar
OFFRATE	Varchar
OFFDES	Varchar
STATUS	Varchar
SOLD	Varchar
DELIVER	Varchar
MISMATCH	Varchar
SERVICE	Varchar
DAMAGE	Varchar
COMPLAINT	Varchar
FEED	Varchar
ADMINACT	Varchar

`

The above table consists of attributes related to Offers. Any complaint towards a product will also be stored in this product. The values obtained from this table will be used in calculating the trust for a product.

PRODUCTS TABLE

ATTRIBUTES	TYPE
PID	Numeric
COMNAME	Varchar
PRONAME	Varchar
WARDATE	Varchar
PROIMAGE	Varchar
PRORATE	Varchar
STATUS	Varchar
ADMINACT	Varchar

The above table consists of different details of the product when the customer views the product. If the seller edits the information the table will be updated.

PURCHASED TABLE

ATTRIBUTES	TYPE
PUR_ID	Numeric
UID	Numeric
UNAME	Varchar
PID	Numeric
COMNAME	Varchar
PRONAME	Varchar
WARDATE	Varchar
PRORATE	Varchar
OFFRATE	Varchar
OFFDES	Varchar
STATUS	Varchar

The above table consists of purchasing details of the product by the customer. Through the PID of the product a product can be uniquely identified

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SELLER TABLE

ATTRIBUTES	TYPE
UID	Numeric
NAME	Varchar
CNAME	Varchar
USERID	Numeric
PASS	Varchar
MOBILE	Varchar
E-MAIL	Varchar
WEB ADD	Varchar
DATE	Varchar
AUTHORIZE	Varchar

The above table stores the details of the seller when they get registered. These details will be further used in validating the user when they login. The status of the seller whether authorized or not will be known through this table.

`

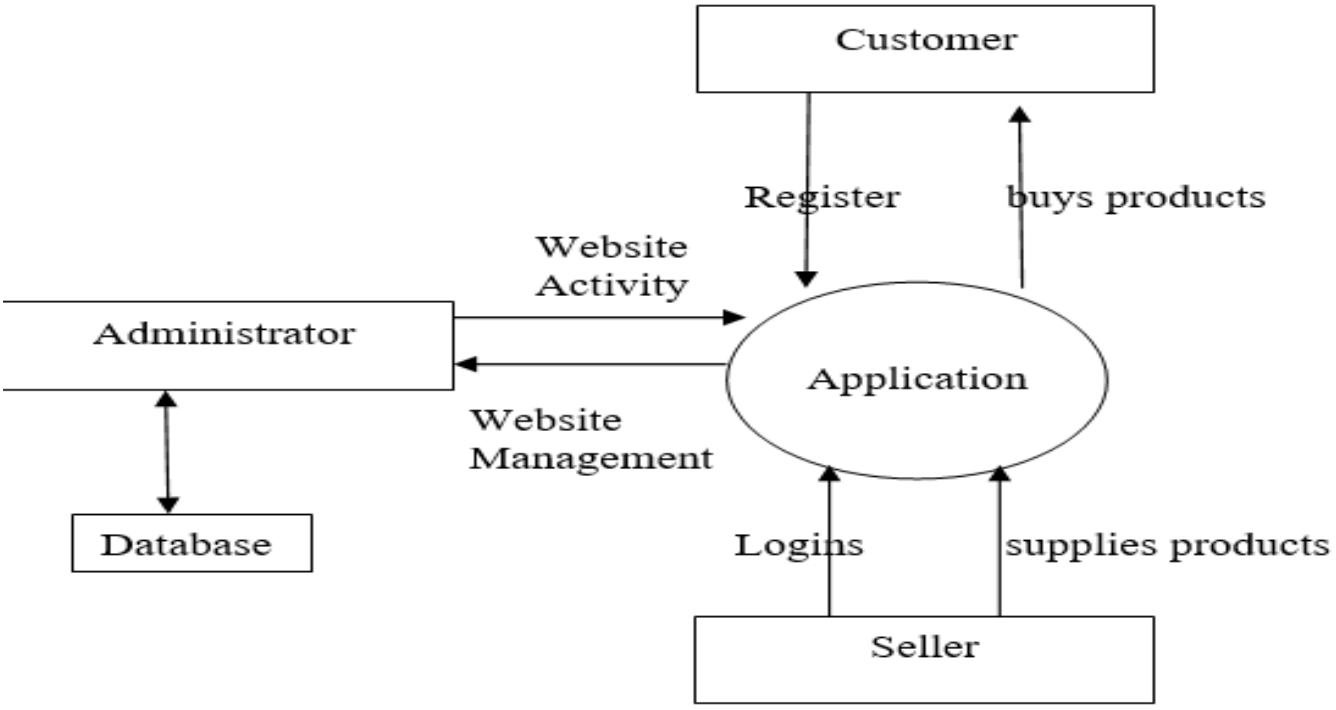
USER TABLE

ATTRIBUTES	TYPE
UID	Numeric
NAME	Varchar
USERID	Numeric
PASS	Varchar
MOBILE	Varchar
EMAIL	Varchar
DATE	Varchar

The above table stores the details of the user when they get registered. These details will be further used in validating the user when they login.

3. DATA FLOW DIAGRAM

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system.



The above dataflow diagram represents the entire system functionality. When the Customer registers to the application he will be able to buy the products and the administrator maintains the website activities by modifying/adding or deleting the sellers. A seller can add/modify/delete the products that are added by him.

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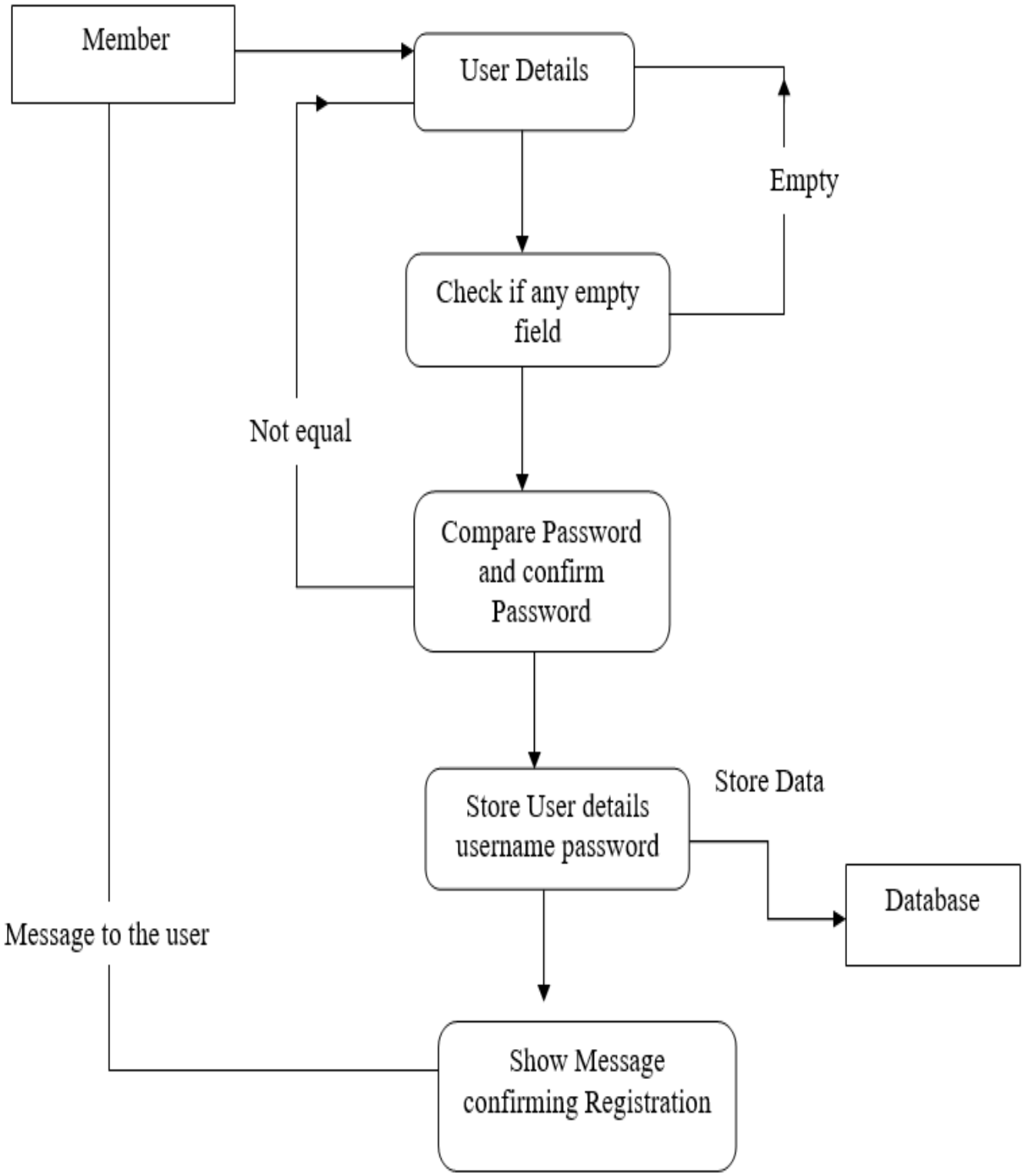
DATA FLOW DIAGRAM FOR REGISTRATION



The above dataflow diagram represents the registration process. A user when wants to register he need to give the required details and when any one of the fields is left empty or forgotten by the user or if the password and confirm password are not equal, the interface will not allow to complete the process until all the fields are properly filled. On successful completion it shows a message confirming user registration.

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SOFTWARE ENVIRONMENT

JAVA TECHNOLOGY

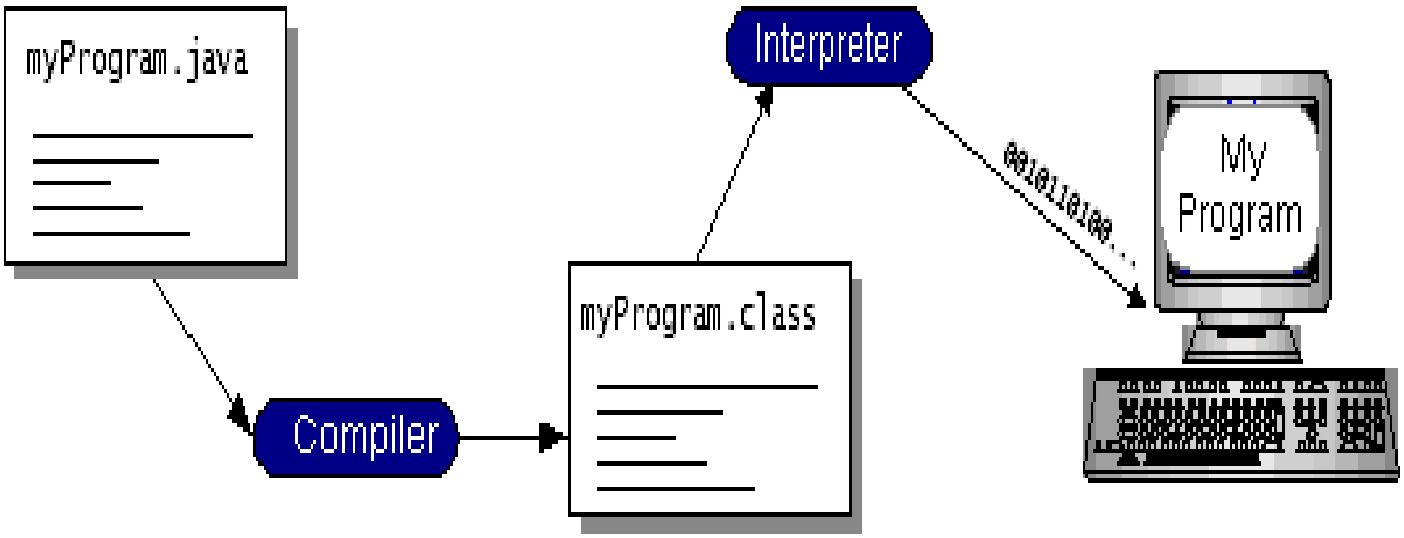
Java technology is both a programming language and a platform.

THE JAVA PROGRAMMING LANGUAGE

The Java programming language is a high-level language that can be characterized by all the following buzzwords:

- Simple
- Architecture neutral
- Object oriented
- Portable
- Distributed
- High performance
- Interpreted
- Multithreaded
- Robust
- Dynamic
- Secure

With most programming languages, you either compile or interpret a program so that you can run it on your computer. The Java programming language is unusual in that a program is both compiled and interpreted. With the compiler, first you translate a program into an intermediate language called Java byte codes —the platform-independent codes interpreted by the interpreter on the Java platform. The interpreter parses and runs each Java byte code instruction on the computer. Compilation happens just once; interpretation occurs each time the program is executed. The following figure illustrates how this works.



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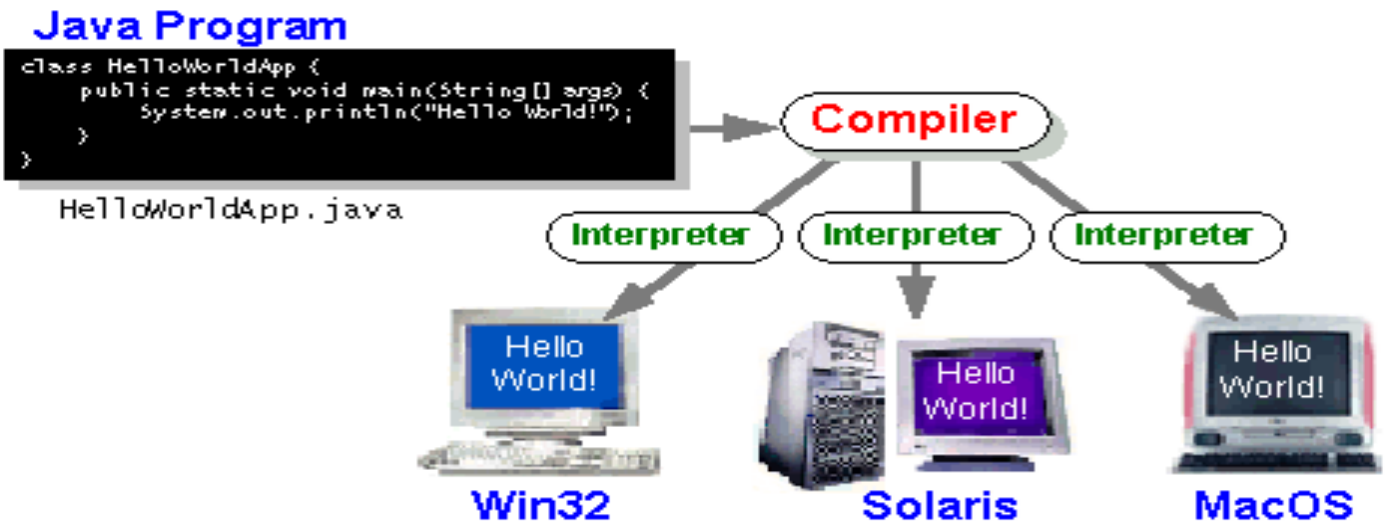
You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a development tool or a Web browser that can run applets, is an implementation of the Java VM. Java byte codes help make "write once, run anywhere" possible. You can compile your program into byte codes on any platform that has a Java compiler. The byte codes can then be run on any implementation of the Java VM. That means that as long as a computer has a Java VM, the same program written in the Java programming language can run on Windows 2000, a Solaris workstation, or on an iMac.



THE JAVA PLATFORM

A platform is the hardware or software environment in which a program runs. We have already mentioned some of the most popular platforms like Windows 2000, Linux, Solaris, and MacOS. Most platforms can be described as a combination of the operating system and hardware. The Java platform differs from most other platforms in that it's a software-only platform that runs on top of other hardware-based platforms.

The Java platform has two components:

- ❖ The Java Virtual Machine (Java VM)
- ❖ The Java Application Programming Interface (Java API)

You have already been introduced to the Java VM. It's the base for the Java platform and is ported onto various hardware-based platforms.

The Java API is a large collection of ready-made software components that provide many useful capabilities, such as graphical user interface (GUI) widgets. The Java API is grouped into libraries of related classes and interfaces; these

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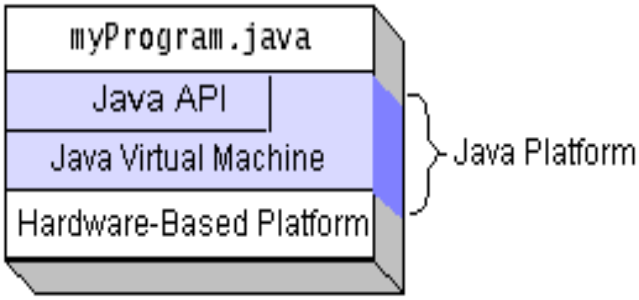
libraries are known as packages. The next section, What Can Java Technology Do? Highlights what functionality some of the packages in the Java API provide.

The following figure depicts a program that's running on the Java platform. As the figure shows, the Java API and the virtual machine insulate the program from the hardware.



Native code is code that after you compile it, the compiled code runs on a specific hardware platform. As a platform-independent environment, the Java platform can be a bit slower than native code. However, smart compilers, well-tuned interpreters, and just-in-time byte code compilers can bring performance close to that of native code without threatening portability.

WHAT CAN JAVA TECHNOLOGY DO?

The most common types of programs written in the Java programming language are applets and applications. If you have surfed the Web, you're probably already familiar with applets. An applet is a program that adheres to certain conventions that allow it to run within a Java-enabled browser.

However, the Java programming language is not just for writing cute, entertaining applets for the Web. The general-purpose, high-level Java programming language is also a powerful software platform. Using the generous API, you can write many types of programs.

An application is a standalone program that runs directly on the Java platform. A special kind of application known as a server serves and supports clients on a network. Examples of servers are Web servers, proxy servers, mail servers, and print servers. Another specialized program is a servlet. A servlet can almost be thought of as an applet that runs on the server side. Java Servlets are a popular choice for building interactive web applications, replacing the use of CGI scripts. Servlets are like applets in that they are runtime extensions of applications. Instead of working in browsers, though, servlets run within Java Web servers, configuring or tailoring the server.

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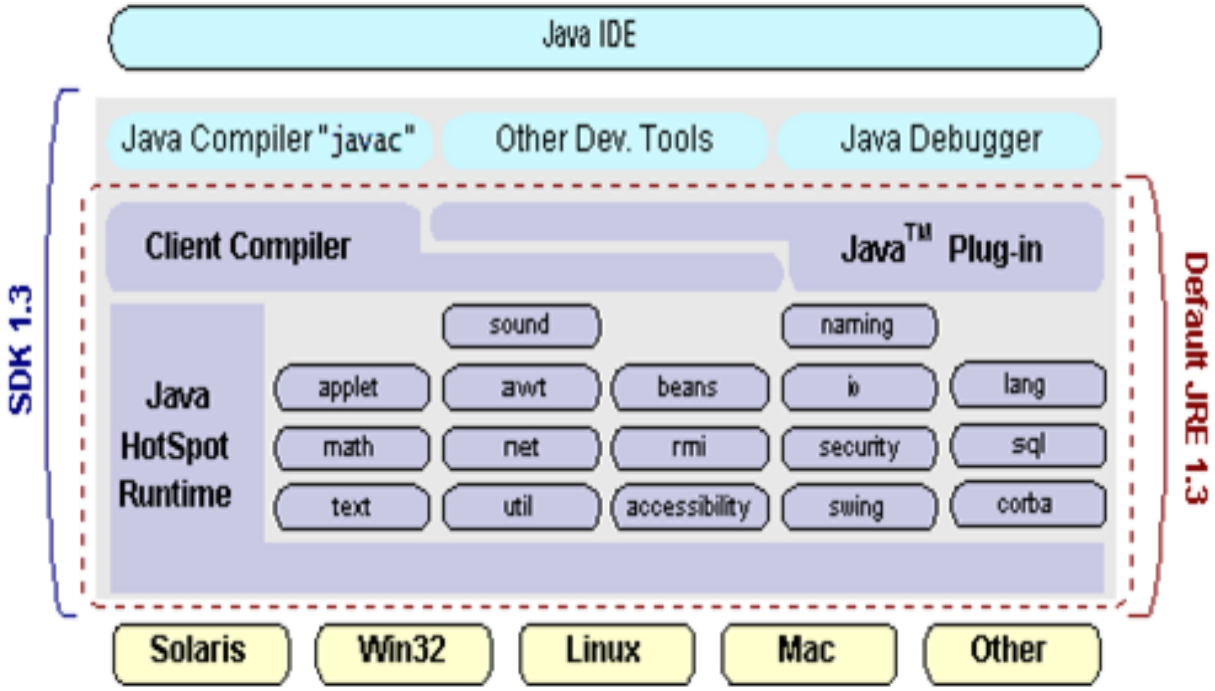
How does the API support all these kinds of programs? It does so with packages of software components that provides a wide range of functionality. Every full implementation of the Java platform gives you the following features:

- The essentials: Objects, strings, threads, numbers, input and output, data structures, system properties, date, and time, and so on.
- Applets: The set of conventions used by applets.
- Networking: URLs, TCP (Transmission Control Protocol), UDP (User Data gram Protocol) sockets, and IP (Internet Protocol) addresses.
- Internationalization: Help for writing programs that can be localized for users worldwide. Programs can automatically adapt to specific locales and be displayed in the appropriate language.
- Security: Both low level and high level, including electronic signatures, public and private key management, access control, and certificates.
- Software components: Known as JavaBeans, can plug into existing component architectures.
- Object serialization: Allows lightweight persistence and communication via Remote Method Invocation (RMI).
- Java Database Connectivity (JDBCTM): Provides uniform access to a wide range of relational databases.

The Java platform also has APIs for 2D and 3D graphics, accessibility, servers, collaboration, telephony, speech, animation, and more. The following figure depicts what is included in the Java 2 SDK.



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HOW WILL JAVA TECHNOLOGY CHANGE MY LIFE?

We can't promise you fame, fortune, or even a job if you learn the Java programming language. Still, it is likely to make your programs better and requires less effort than other languages. We believe that Java technology will help you do the following:

- Get started quickly: Although the Java programming language is a powerful object-oriented language, it's easy to learn, especially for programmers already familiar with C or C++.
- Write less code: Comparisons of program metrics (class counts, method counts, and so on) suggest that a program written in the Java programming language can be four times smaller than the same program in C++.
- Write better code: The Java programming language encourages good coding practices, and its garbage collection helps you avoid memory leaks. Its object orientation, its JavaBeans component architecture, and its wide-ranging, easily extendible API let you reuse other people's tested code and introduce fewer bugs.
- Develop programs more quickly: Your development time may be as much as twice as fast versus writing the same program in C++. Why? You write fewer lines of code and it is a simpler programming language than C++.
- Avoid platform dependencies with 100% Pure Java: You can keep your program portable by avoiding the use of libraries written in other languages. The 100% Pure JavaTM Product Certification Program has a repository of historical process manuals, white papers, brochures, and similar materials online.
- Write once, run anywhere: Because 100% Pure Java programs are compiled into machine-independent byte codes, they run consistently on any Java platform.
- Distribute software more easily: You can upgrade applets easily from a central server. Applets take advantage of the feature of allowing new classes to be loaded "on the fly," without recompiling the entire program.

ODBC

Microsoft Open Database Connectivity (ODBC) is a standard programming interface for application developers and database systems providers. Before

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ODBC became a de facto standard for Windows programs to interface with database systems, programmers had to use proprietary languages for each database they wanted to connect to. Now, ODBC has made the choice of the database system almost irrelevant from a coding perspective, which is as it should be. Application developers have much more important things to worry about than the syntax that is needed to port their program from one database to another when business needs suddenly change.

Through the ODBC Administrator in Control Panel, you can specify the database that is associated with a data source that an ODBC application program is written to use. Think of an ODBC data source as a door with a name on it. Each door will lead you to a particular database. For example, the data source named Sales Figures might be a SQL Server database, whereas the Accounts Payable data source could refer to an Access database. The physical database referred to by a data source can reside anywhere on the LAN.

The ODBC system files are not installed on your system by Windows 95. Rather, they are installed when you setup a separate database application, such as SQL Server Client or Visual Basic 4.0. When the ODBC icon is installed in Control Panel, it uses a file called ODBCINST.DLL. It is also possible to administer your ODBC data sources through a stand-alone program called ODBCADM.EXE. There is a 16-bit and a 32-bit version of this program and each maintains a separate list of ODBC data sources.

From a programming perspective, the beauty of ODBC is that the application can be written to use the same set of function calls to interface with any data source, regardless of the database vendor. The source code of the application doesn’t change whether it talks to Oracle or SQL Server. We only mention these two as an example. There are ODBC drivers available for several dozen popular database systems. Even Excel spreadsheets and plain text files can be turned into data sources. The operating system uses the Registry information written by ODBC Administrator to determine which low-level ODBC drivers are needed to talk to the data source (such as the interface to Oracle or SQL Server). The loading of the ODBC drivers is transparent to the ODBC application program. In a client/server environment, the ODBC API even handles many of the network issues for the application programmer.

The advantages of this scheme are so numerous that you are probably thinking there must be some catch. The only disadvantage of ODBC is that it isn’t as

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efficient as talking directly to the native database interface. ODBC has had many detractors make the charge that it is too slow. Microsoft has always claimed that the critical factor in performance is the quality of the driver software that is used. In our humble opinion, this is true. The availability of good ODBC drivers has improved a great deal recently. And anyway, the criticism about performance is somewhat analogous to those who said that compilers would never match the speed of pure assembly language. Maybe not, but the compiler (or ODBC) gives you the opportunity to write cleaner programs, which means you finish sooner. Meanwhile, computers get faster every year.

JDBC

To set an independent database standard API for Java; Sun Microsystems developed Java Database Connectivity, or JDBC. JDBC offers a generic SQL database access mechanism that provides a consistent interface to a variety of RDBMSs. This consistent interface is achieved through the use of "plug-in" database connectivity modules, or drivers. If a database vendor wishes to have JDBC support, he or she must provide the driver for each platform that the database and Java run on.

To gain a wider acceptance of JDBC, Sun based JDBC's framework on ODBC. As you discovered earlier in this chapter, ODBC has widespread support on a variety of platforms. Basing JDBC on ODBC will allow vendors to bring JDBC drivers to market much faster than developing a completely new connectivity solution.

JDBC was announced in March of 1996. It was released for a 90 day public review that ended June 8, 1996. Because of user input, the final JDBC v1.0 specification was released soon after.

The remainder of this section will cover enough information about JDBC for you to know what it is about and how to use it effectively. This is by no means a complete overview of JDBC. That would fill an entire book.

JDBC GOALS

Few software packages are designed without goals in mind. JDBC is one that, because of its many goals, drove the development of the API. These goals, in conjunction with early reviewer feedback, have finalized the JDBC class library into a solid framework for building database applications in Java.

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The goals that were set for JDBC are important. They will give you some insight as to why certain classes and functionalities behave the way they do. The eight design goals for JDBC are as follows:

- SQL Level API

The designers felt that their main goal was to define a SQL interface for Java. Although not the lowest database interface level possible, it is at a low enough level for higher-level tools and APIs to be created. Conversely, it is at a high enough level for application programmers to use it confidently. Attaining this goal allows for future tool vendors to "generate" JDBC code and to hide many of JDBC's complexities from the end user.

- SQL Conformance

SQL syntax varies as you move from database vendor to database vendor. In an effort to support a wide variety of vendors, JDBC will allow any query statement to be passed through it to the underlying database driver. This allows the connectivity module to handle non-standard functionality in a manner that is suitable for its users.

- JDBC must be implemental on top of common database interfaces

The JDBC SQL API must "sit" on top of other common SQL level APIs. This goal allows JDBC to use existing ODBC level drivers by the use of a software interface. This interface would translate JDBC calls to ODBC and vice versa.

- Provide a Java interface that is consistent with the rest of the Java system

Because of Java's acceptance in the user community thus far, the designers feel that they should not stray from the current design of the core Java system.

- Keep it simple

This goal probably appears in all software design goal listings. JDBC is no exception. Sun felt that the design of JDBC should be very simple, allowing for only one method of completing a task per mechanism. Allowing duplicate functionality only serves to confuse the users of the API.

- Use strong, static typing wherever possible

Strong typing allows for more error checking to be done at compile time; also, less error appear at runtime.

- Keep the common cases simple

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Because often, the usual SQL calls used by the programmer are simple SELECT's, INSERT's, DELETE's and UPDATE's, these queries should be simple to perform with JDBC. However, more complex SQL statements should also be possible.

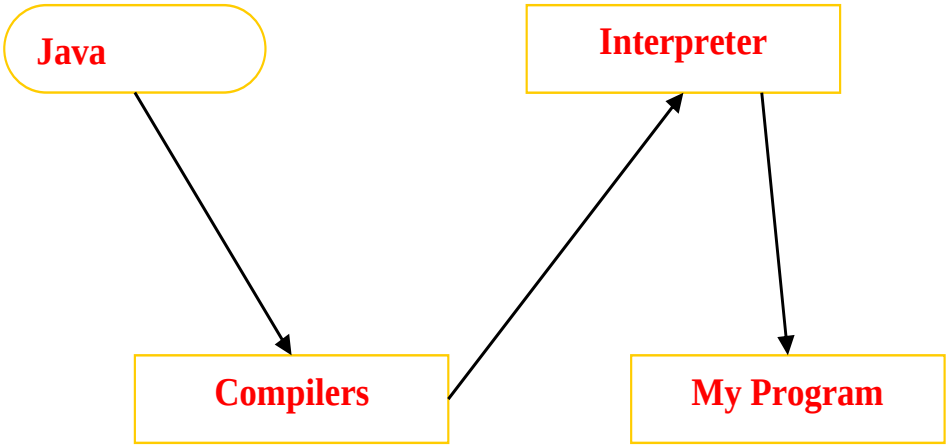
Finally, we decided to proceed the implementation using Java Networking.

And for dynamically updating the cache table we go for MS Access database.

Java has two things: a programming language and a platform. Java is a high-level programming language that is all the following

Simple	Architectural neutral
Object oriented	portable
Distributed	High Performance
Interpreted	multithreaded
robust	dynamic
Secure	

Java is also unusual in that each Java program is both compiled and interpreted. With a compile you translate a Java program into an intermediate language called Java byte codes the platform-independent code instruction is passed and run on the computer. Compilation happens just once; interpretation occurs each time the program is executed. The figure illustrates how this works.



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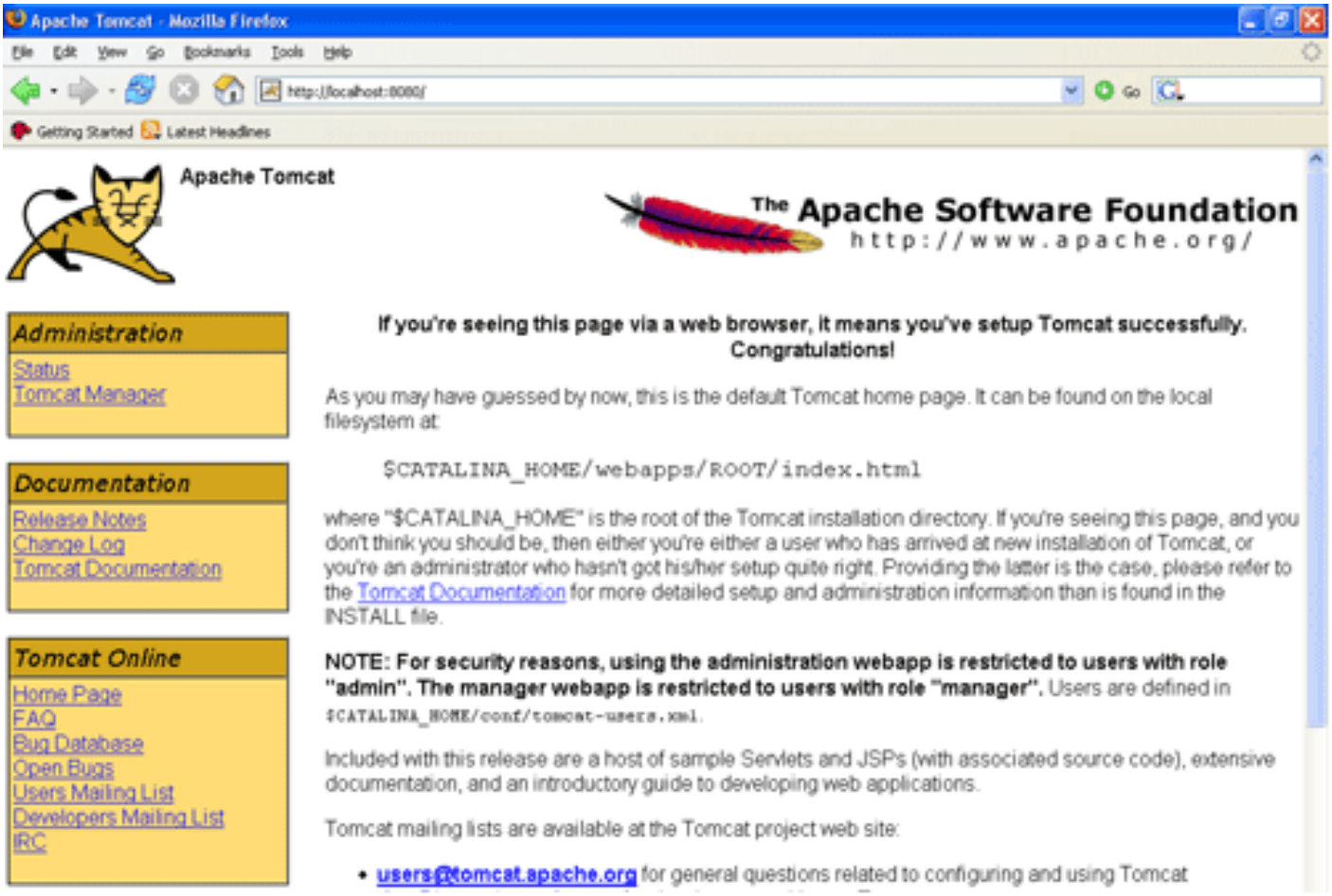
Java byte codes help make “write once, run anywhere” possible. You can compile your Java program into byte codes on my platform that has a Java compiler. The byte codes can then be run any implementation of the Java VM. For example, the same Java program can run Windows NT, Solaris, and Macintosh.

TOMCAT 6.0 WEB SERVER

Tomcat is an open-source web server developed by Apache Group. Apache Tomcat is the servlet container that is used in the official Reference Implementation for the Java Servlet and Java Server Pages technologies. The Java Servlet and Java Server Pages specifications are developed by Sun under the Java Community Process. Web Servers like Apache Tomcat support only web components while an application server supports web components as well as business components (BEAs WebLogic, is one of the popular application server).To develop a web application with jsp/servlet install any web server like JRun, Tomcat etc to run your application.



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IMPLEMENTATIONS

Implementation is the stage of the project when the theoretical design is turned out into a working system. Thus, it can be the most critical stage in achieving a successful new system and in giving the user, confidence that the new system will work and be effective. The implementation stage involves careful planning, investigation of the existing system and its constraints on implementation, designing of methods to achieve changeover and evaluation of changeover methods.

MODULES AND THEIR FUNCTIONALITIES

The system has been classified into the following modules after a careful analysis,

1. Customer Module
2. Seller Module
3. Administrative Module
4. Complaint Filing
5. Fraud churn

CUSTOMER MODULE

A customer is one of the users who wish to shop online. For this purpose, the customer will be provided with a personal account through registration. After successful registration, he will be provided with a gallery of different products from different sellers which include the product name, price, sellers' name etc. While buying a product a customer can view the percent of trustworthiness towards the product given by other users. After purchasing, a customer can also file complaint on that product where he feels uncomfortable provided with some options like

- Products purchased by the buyer are not delivered by the seller.
- The delivered products do not match the descriptions that were posted by sellers.
- Malicious sellers may even post non-existing items with false description to deceive buyers
- General feedback as a complaint

SELLER MODULE

The seller module includes different sellers who wish to sell their products. The seller needs to be approved by administrator after a seller submits his

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registration. A seller can add or delete or modify information about different items.

The different functionalities for seller are

- Can add a new a product
- Can delete a product
- Can place new offers to the product
- Can modify information related to the product such as price, basic information etc...

ADMIN MODULE

The administrative module includes an admin who acts as an intermediator between seller and the customer. An Adminis responsible to maintain the website information giving a trust to the customers. When a complaint is filed in the customer module, the admin takes the final decision whether to ban the product. If the admin feels all the products from seller mostly are not trusted he can also remove the seller and his related products**.**

COMPLAINT FILLING

Buyers can file complaints to claim loss if they are recently deceived by fraudulent sellers. The Administrator views the various types of complaints and the percentage of various type complaints. The complaints values of a products increase some threshold value the administrator set the trust ability of the product as Untrusted or banned. If the products set as banned, the user cannot view the products in the website.

FRAUD CHURN

In this module admin takes the decision whether to continue the seller to sell the products or not. When some products are labelled as fraud by human experts, it is very likely that the seller is not trustable and the products too. Hence all the items submitted by the same seller are labelled as fraud too. So, the fraudulent seller along with his/her cases will be removed from the website immediately once detected.

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TESTING

Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides way to check the functionalities of the components, assemblies and or a finished product. It is the process of exercising the software with the intent of ensuring that the software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses specific testing requirements.

TESTING OBJECTIVES

- Testing is a process of executing a program with the intent of finding an error.
- A good test has a high probability of finding an as yet undiscovered error.
- A successful test is one uncovers an as yet undiscovered error.

TYPES OF TESTS

a. System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

b. Whitebox testing

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

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c. Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements documents. It is a testing in which the software under test is treated, as a Black Box.

d. Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

All field entries must work properly.

Pages must be activated from the identified link.

The entry screen, messages and responses must not be delayed.

Features to be tested

Verify that the entries are of the correct format

No duplicate entries should be allowed

All links should take the user to the correct page.

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e. Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g., components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

f. Acceptance testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements. Test Results: All the test cases mentioned above passed successfully. No defects encountered.

`

TEST CASE ANALYSICS

Some of the test cases and their expected results are:

TEST CASE ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS PASS/FAIL
1.	Type Wrong Username and Password for any user	An Error message has to be displayed. It should prompt for password	An error message is displayed prompting wrong password	P
2.	Type correct Username and password	Home page should displayed	Home page is displayed	P
3.	Any field regarding to product adding is not given	Should prompt for that specific field	Prompting for the empty field	P
4.	Any field left blank during registration	Should prompt for completion of registration	Prompting to enter the specific field	P
5.	Click logout	Should come to the Main page	Main page is displayed if logout is clicked	P
6.	Verification of admin name and password	In Correct owner name, password, system should not login	System did not log in	P
7.	Verification of Seller name and password	In Correct Seller name, In correct password, system should not login	System did not log in	P
8.	Verification of Seller upload the products in the application	In correct product name, product image, product description and offer price	The product did not uploaded	P
9.	Verification of user purchase the products	User can select the product and based on offers he can purchase the product	The user can able to purchase the product and product are maintained in shopping cart	P
10.	Verification of users complaints	user complaint details need to updated in the database and details are forwarded to the admin	user complaints details are updated and details are forwarded to the admin	P

`

11.	Verification of admin getting complaint details	The data is reading from server and data is presented in the admin page	The data is reading from server and data is presented in the admin page, once he verified complaint details move to status to trusted or untrusted	P
12.	Verification of admin block the product	The product need to removed from the application	The product removed from the application	P

`

CONCLUSION

Since the emergence of the World Wide Web (WWW), electronic commerce, commonly known as e-commerce, has become more and more popular, websites benefits everyone in terms of convenience and profitability . The traditional online shopping business model allows sellers to sell a product or service at a preset price, where buyers can choose to purchase find it to be a good deal but we build online model for fraud product detection while concentrating on customer needs.In this proposed system we provide the responsibility of selling the trustful products by the website itself managed by the admin. So, when a customer wishes to buy a product he will get an idea about the product to how much extent he can believe in that product. If he has faced any problem, he can make others aware of that product by complaining about the product. This model though it cannot be the ideal way of detecting frauds but it can do the maximum extent in detecting the sellers selling the fraud products.

The true online shopping is that which discovers each customer’s known interests and needs on an individual level and gives a much more powerful platform from which to optimize content and offers, a vital key to long-term brand engagement and loyalty.

`

FURTHER ENHANCEMENTS

Regarding to future work, one direction is to include the adjustment of the selection bias in the online model training process. It has been proven to be very effective for offline models. The main idea there is to assume all the unlabelled samples have response equal to 0 with a very small weight. Since the unlabelled samples are obtained from an effective moderation system, it is reasonable to assume that with high probabilities they are non-fraud.

This can be easily extended to too many other applications, such as web spam detection, content optimization and so forth websites that delivers highly personalized and trusted experiences top the traffic and revenue rankings across the globe.

Web spam has been an important problem affecting both the consumers and web service providers since the invention of World Wide Web. So we can attempt to build a spam detection system for classification of websites as spam or non-spam. Here we try to explore if the spam web-sites follow certain pattern in terms of the links they are out linked/in linked to or in terms of contents of such websites. For this, we use various features based on the link graph or the contents of only the host pages. The benefit of host-based labelling instead of individual page based labelling is that we can cover a larger number of websites to build the model. We define spam city as the probability with which a page can be classified as spam (0 for non-spam page and 1 for spam page).

`

APPENDIX A: BIBLIOGRAPHY

- D. Chau and C. Faloutsos, “Fraud detection in electronic auction”. In European Web Mining Forum (EWMF 2005), page 87.
- Liang Zhang Jie Yang Belle Tseng, “Online Modelling of Proactive Moderation System for Auction Fraud Detection”, Yahoo! Labs 701 First Ave Sunnyvale, USA@yahoo-inc.com
- Magdalini Eirinaki and Michalis Vazirgiannis, “Web Personalization” Athens University of Economics and Business. Department of Informatics.
- W3Schools Online Web Tutorials.

`

APPENDIX B: SAMPLE SOURCE CODE

Main page.jsp

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
<head>
<meta http-equiv="content-type" content="text/html; charset=iso-8859-1" />
<title>Auction Fraud</title>
<link href="style.css" rel="stylesheet" media="all" type="text/css" />
</head>
<body>
<div id="wrapper">
<div id="container">
	<div id="header">
	<div
id="logo"><br><br><br>       &n
bsp;   <strong><font color="#FFFFFF" size="+2"
face="Georgia, Times New Roman, Times, serif"> Detection of Fraudulent
Sellers in Online Marketplaces <br>
          
           
         
      </font></strong></div>
	</div>
	<div id="navbar">
	<ul>
		<li><a href="index.html" class="active">Home</a></li>
	</ul>
	</div>
```

`

<table height="350" align="center" width="700">

<tr bgcolor="#CC3300">

<td width="610" bgcolor="#FBF7E1" valign="top">

           <font color="#FF0000" size="+1" face="Courier New">

Detection of Fraudulent Sellers</font>

 The E-business sector is rapidly evolving and the needs for web market places that anticipate the needs of the customers and the trust towards a product are equally more evident than ever. While people are enjoying the benefits from online trading, criminals are also taking advantages to conduct fraudulent activities against honest parties to obtain illegal profits. Therefore the requirement for predicting user needs and trust providence in order to improve the usability and user retention of a website can be addressed by personalizing and using a fraud product detection system.Hence fraud-detection systems are commonly needed to be applied to detect and prevent such illegal or untrusted products. In this, we propose an online model framework which takes online feature selection, coefficient bounds from human knowledge and multiple instances learning into account simultaneously. By empirical experiments on a real-world we show that this model can potentially meet user needs, calculate the trust for a product and significantly reduce customer complaints.

</td>

<td width="147" bgcolor="#F3ECC2"><table>

<tr>

<td align="center"><font color="#FF0000" size="+1" face="Georgia, Times New Roman, Times, serif"><img src="images/reg.png" width="35" height="35">Registration</font></td>

```
`
        </tr><tr>
<td align="center"><font face="Comic Sans MS" size="3" class="big"><a
href="seller_signup.jsp">Seller</a></font></td>
        </tr><tr>
<td align="center"><font face="Comic Sans MS" size="3" class="big"><a
href="user_signup.jsp">User</a></font></td>
</tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>
<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr>
<td align="center"><font color="#FF0000" size="+1" face="Georgia, Times
New Roman, Times, serif"><strong><img src="images/log1.png" width="35"
height="35"><br>
                Login</strong></font></td>
        </tr><tr>
<td align="center"><font face="Comic Sans MS" size="+1" class="big"><a
href="seller_log.jsp">Seller</a></font></td>
        </tr><tr>
<td align="center"><font face="Comic Sans MS" size="+1" class="big"><a
href="user_log.jsp">User</a></font></td>
        </tr><tr>
<td align="center"><font face="Comic Sans MS" size="+1" class="big"><a
href="admin_log.jsp">Admin</a></font></td>
</tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>
<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>
</table></td>
                </tr>
                </table>
                </div>
                <div id="columns-wrapper">
                </div>
```

```
`
            </div>
            <div id="footer">
            <div id="footer-right"> </div>
            <div id="footer-left"> </div>
                <br>
                <br>
            </div>
            </div>
            </div>
</body>
</html>
```

ProductDispaly.jsp

```
<%@ page  import="java.sql.*" import="databaseconnection.*"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
<head>
<meta http-equiv="content-type" content="text/html; charset=iso-8859-1" />
<title>Auction Fraud</title>
<link href="style.css" rel="stylesheet" media="all" type="text/css" />
</head>
<body>
<%
String name=(String)session.getAttribute("name");
String u=(String)session.getAttribute("u");
```

```
`
System.out.println(u);
%>
<div id="wrapper">
        <div id="container">
        <div id="header"><div id="logo"><br><br><br>
           
<strong><font color="#FFFFFF" size="+2" face="Georgia, Times New Roman,
Times, serif">
Detection of Fraudulent Sellers in Online Marketplaces
<br>          
           
         
      </font></strong></div></div>
<div id="navbar">
<ul>
        <li><a href="user_home.jsp" >Home</a></li>
        <li><a href="my_products.jsp" class="active">My Products</a></li>
        <li><a href="index.html">Logout</a></li>
           
           
 
           
          

    <font color="#333366" face="Georgia, Times New
Roman, Times, serif" size="+1"><strong>welcome:</strong></font>
 <font color="#FF0000" face="Georgia, Times New Roman, Times, serif"
size="+1"><strong><%=name%></strong></font>
</ul>
</div>
<div id="main">
```

```
`
<div id="intro">
<div id="text"></div>
<table height="350" align="center" width="700">
<tr bgcolor="#CC3300">
<td width="610" bgcolor="#FBF7E1" ><strong><font color="#FF0000"
size="+1" face="Georgia, Times New Roman, Times, serif"><em>My
            Products</em></font></strong><br><br><form name="f" action="#"
method="post" onsubmit="return valid()">
<table bgcolor="#FFFFFF" width="700" border="0">
<tr bgcolor="#E4E4F1">
<td align="center"><font color="#110022"><strong>Purchase
                ID</strong></font></td>
<td align="center"><font color="#110022"><strong>Company
                Name</strong></font></td>
<td align="center"><font color="#110022"><strong>Product
ID</strong></font></td>
<td align="center"><font color="#110022"><strong>Product
Name</strong></font></td>
<td align="center"><font color="#110022"><strong>Warrenty
                date</strong></font></td>
<td align="center"><font color="#110022"><strong>Product
Rate</strong></font></td>
<td align="center"><font
color="#110022"><strong>Description</strong></font></td>
<td align="center"><font
color="#110022"><strong>Complaint</strong></font></a></td>
</tr>
<%
```

```
`
String
pname=null,pid=null,cname=null,purid=null,orate=null,des=null,wdate=null;
ResultSet rs=null;
try
{
        Connection con = databasecon.getconnection();
        Statement st = con.createStatement();
   String qry="select * from purchased where uname='"+name+"' &&
uid='"+u+"'";
        rs =st.executeQuery(qry);
        while(rs.next())
        {
        purid=rs.getString("pur_id");
        cname=rs.getString("comname");
        pid=rs.getString("pid");
        pname=rs.getString("proname");
        wdate=rs.getString("wardate");
        orate=rs.getString("offrate");
        des=rs.getString("offdes");
%>
<tr bgcolor="#FFFFCC">
<td align="center"><strong><font color="#FF0000"><%=purid%>
</font></strong></td><td align="center"><strong><font
color="#6300C6"><%=cname%>
</font></strong></td>
<td align="center"><strong><font color="#6300C6"><%=pid%>
</font></strong></td>
```

```
`
<td align="center"><strong><font
color="#6300C6"><%=pname%></font></strong></td>

<td align="center"><strong><font
color="#6300C6"><%=wdate%></font></strong></td>

<td align="center"><strong><font
color="#6300C6"><%=orate%></font></strong></td>

<td align="center"><strong><font
color="#6300C6"><%=des%></font></strong></td>

<td align="center"><strong><font color="#6300C6"><a
href="user_complaint.jsp?<%=pid%>"><font
color="#FF0000"><strong>Complaint</strong></font></a></font></strong></t
d>

</tr>

<%

}        }

         catch(Exception e1)

         {

           out.println(e1.getMessage());

         }

 %>

</table>

</form></td>

</tr>

</table>

</div>

<div id="columns-wrapper">

</div>

</div>

<div id="footer">
```

```
`
<div id="footer-right"> </div>
<div id="footer-left"> </div>
<br><br>
</div>
</div>
</div>
</body>
</html>
```

Trust.jsp

```
<%@ page  import="java.sql.*" import="databaseconnection.*"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
<head>
<meta http-equiv="content-type" content="text/html; charset=iso-8859-1" />
<title>Auction Fraud</title>
<link href="style.css" rel="stylesheet" media="all" type="text/css" />
<style type="text/css">
#bg
{
background-color:white;
width:50px;
height:100px;
}
</style>
</head>
<body>
```

```
`
<div id="wrapper">
<div id="container">
        <div id="header">
        <div id="logo"><br>
<br><br>
           
<strong><font color="#FFFFFF" size="+2" face="Georgia, Times New Roman,
Times, serif">
        Detection of Fraudulent Sellers in Online Marketplaces <br>
                  
           
         
       </font></strong></div>
        </div>
<div id="navbar">
        <ul>
                <li><a href="admin_home.jsp" class="active">Home</a></li>
</ul>
        </div>
<div id="main">
        <div id="intro">
<div id="text"></div>
        <table height="350" align="center" width="700">
        <tr bgcolor="#CC3300">
<td width="610" bgcolor="#FBF7E1" align="center"><strong><font
color="#FF3300" size="+1" face="Georgia, Times New Roman, Times,
serif"><br>
<br>
            Product Survey Status</font></strong><br><br><br>
```

```
`
<%
                    String tpid=request.getQueryString();
        String sold=null, del=null,
miss=null,serv=null,dam=null,pname=null,cname=null;
ResultSet rs=null;
try
{
                    Connection con = databasecon.getconnection();
                    Statement st = con.createStatement();
   String qry="select * from offers where pid='"+tpid+"'";
                    rs =st.executeQuery(qry);
                    while(rs.next())
                    {
                     pname=rs.getString("proname");
                    cname=rs.getString("comname");
                    sold=rs.getString("sold");
                    del=rs.getString("deliver");
                    miss=rs.getString("missmatch");
                    serv =rs.getString("service");
                    dam =rs.getString("damage");
        }

            int sold1=Integer.parseInt(sold);
            int del1=Integer.parseInt(del);
              int miss1=Integer.parseInt(miss);
            int serv1=Integer.parseInt(serv);
             int dam1=Integer.parseInt(dam);
```

```
`
                int sum=del1+miss1+serv1+dam1;
     Double sum1=sum/((0.01)*(sold1));
//System.out.println(sum1);
                double t=50.0;
                Double tru=100-sum1;
            %>
        <fieldset>
        <br>
        <br>
        <table width="513" height="394" cellpadding="5" cellspacing="5">
        <tr>
<td width="250"><font face="Georgia, Times New Roman, Times, serif"
color="#330033" size="+1"><strong>Product
                ID</strong></font></td>
<td width="162"><font face="Courier New, Courier, mono" size="+2"
color="#FF3300"><strong><%=tpid%></strong></font></td>
        </tr><tr>
<td><font face="Georgia, Times New Roman, Times, serif" color="#330033"
size="+1"><strong>
                Product Name</strong></font></td>
<td><font face="Georgia, Times New Roman, Times, serif" color="#FF0000"
size="+1"><%=pname%></font></td>
        </tr>
<tr>
<td><font face="Georgia, Times New Roman, Times, serif" color="#330033"
size="+1"><strong>Company
                Name</strong></font></td>
<td><font face="Georgia, Times New Roman, Times, serif" color="#FF0000"
size="+1"><%=cname%></font></td>
```

```
`
</tr><tr>
<td><font face="Georgia, Times New Roman, Times, serif" color="#330033" size="+1"><strong>Number of Sold
</strong></font></td>
<td><font face="Georgia, Times New Roman, Times, serif" color="#FF0000" size="+1"><%=sold%></font></td>
</tr>
<tr>
<td><strong><font face="Georgia, Times New Roman, Times, serif" color="#330033" size="+1">Complaints</font></strong></td>
<td><img src="images/sca1.jpg" width="50" height="100"><img src="images/bar_red1.jpg" width="50" height="<%=sum1%>">
<br><br><font size="+1" color="#6633FF"><%=sum1%></font> <font size="+1" color="#FF0000"><strong>%</strong></font></td>
</tr>
<tr><tr>
<td><font face="Georgia, Times New Roman, Times, serif" color="#330033" size="+1"><strong>Trustability</strong></font></td>
<td ><img src="images/sca1.jpg" width="50" height="100"><img src="images/bar_gree.jpg" width="50" height="<%=tru%>">
<br><br><font size="+1" color="#6633FF"><%=tru%></font> <font size="+1" color="#FF0000"><strong>%</strong></font></td>
</tr>
<tr></tr>
<tr></tr>
<tr>
<td><a href="admin_home.jsp"><strong><font size="+1" face="Courier New" color="#FF0000">Back</font></strong></a></td>
<td><a href="more_det.jsp?<%=tpid%>"><strong><font size="+1" face="Courier New" color="#FF0000">More
```

```
`
                    Details</font></strong></a></td>
</tr>
</table>
<br>
<br>
</fieldset>
<%
            }
           catch(Exception e1)
           {
             out.println(e1.getMessage());
           }
 %>
</td>
</tr>
</table>
</div>
<div id="columns-wrapper">
</div>
</div>
<div id="footer">
<div id="footer-right"> </div>
<div id="footer-left"> </div>
<br>
<br>
</div>
</div>
```

```
`
</div>
</body>
</html>
```

UserComplaint.jsp

```
<%@ page  import="java.sql.*" import="databaseconnection.*"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
<head>
<meta http-equiv="content-type" content="text/html; charset=iso-8859-1" />
<title>online shopping</title>
<script type="text/javascript">
function valid()
{
if(document.f.op[0].checked==false&&document.f.op[1].checked==false&&doc
ument.f.op[2].checked==false&&document.f.op[3].checked==false)
{
alert("select Complaint");
return false;
}
var a=document.f.com1.value;
if(a==="")
{
alert("enter Complaint");
document.f.com1.focus();
return false;
}
```

```
`
}
</script>
<link href="style.css" rel="stylesheet" media="all" type="text/css" />
</head>
<body>
<%
String name=(String)session.getAttribute("name");
String pid1=request.getQueryString();
session.setAttribute("pid1",pid1);
%>
<div id="wrapper">
<div id="container">
        <div id="header">
        <div id="logo"><br>
        <br>
        <br>           <strong><font color="#FFFFFF" size="+2" face="Georgia, Times New Roman, Times, serif">
       Detection of Fraudulent Sellers in Online Marketplaces <br>
           
```

```
`
        <li><a href="index.html">Logout</a></li>
        <li><a href="#">Link</a></li>
        <li><a href="#">Link</a></li>
        <li><a href="#">Link</a></li>-->
           
           
 
nbsp;          &
nbsp;         
    <font color="#333366" face="Georgia, Times New
Roman, Times, serif" size="+1"><strong>welcome:</strong></font>
 <font color="#FF0000" face="Georgia, Times New Roman, Times, serif"
size="+1"><strong><%=name%></strong></font>
        </ul>
        </div>
        <div id="main">
        <div id="intro">
<div id="text"></div>
        <table height="350" align="center" width="700">
        <tr bgcolor="#CC3300">
<td width="300" bgcolor="#FBF7E1" valign="top"><form name="f"
action="user_com_insert.jsp" method="post" onsubmit="return valid()">
<fieldset>
<legend><font color="#FF0000" size="+2" face="Courier
New"><strong><em>Complaint</em></strong></font></legend>
<table width="271" cellpadding="10" cellspacing="5">
<tr>
<td colspan="2" align="center"><font size="2"><b>
<%
```

```
`
            String message=request.getParameter("message");
 if(message!=null && message.equalsIgnoreCase("success"))
                        {
  out.println("<font color='red'><blink>Complaint Registered !</blink></font>");
                        }
  %>
</b></font></td>
</tr>
<tr>
<td><strong><font color="#CC0000" size="+1" face="Georgia, Times New Roman, Times, serif">complaint about</font></strong></td>
<td><input type="radio" name="op" value="deliver" ><strong><font color="#330000">Not Delivered</font></strong><br><br><input type="radio" name="op" value="missmatch">
<strong><font color="#330000">Product Missmatch</font></strong><br>
<br><input type="radio" name="op" value="service"><strong><font color="#330000">Poor Service</font></strong><br><br><input type="radio" name="op" value="damage">
<strong><font color="#330000">Product Damaged</font></strong><br>
<br></td>
</tr>
<tr>
<td><strong><font color="#CC0000" size="+1" face="Georgia, Times New Roman, Times, serif">Enter Complaint </font></strong></td>
<td><textarea name="com1" cols="12"></textarea></td>
</tr>
<tr>
<td><input type="reset" name="r" value="clear" class="btn"></td>
<td><input type="submit" name="s" value="submit" class="btn"></td>
```

```
`
</tr>
</table>
</fieldset>
</form></td>
<td width="100" bgcolor="#FBF7E1" align="center"><img
src="images/comp.png" height="150" width="150"></td>
        </td>
</tr>
</table>
</div>
<div id="columns-wrapper">
</div>
</div>
<div id="footer">
<div id="footer-right"> </div>
<div id="footer-left"> </div>
<br><br>
</div>
        </div>
</div>
</body>
</html>
```

Authorize.jsp

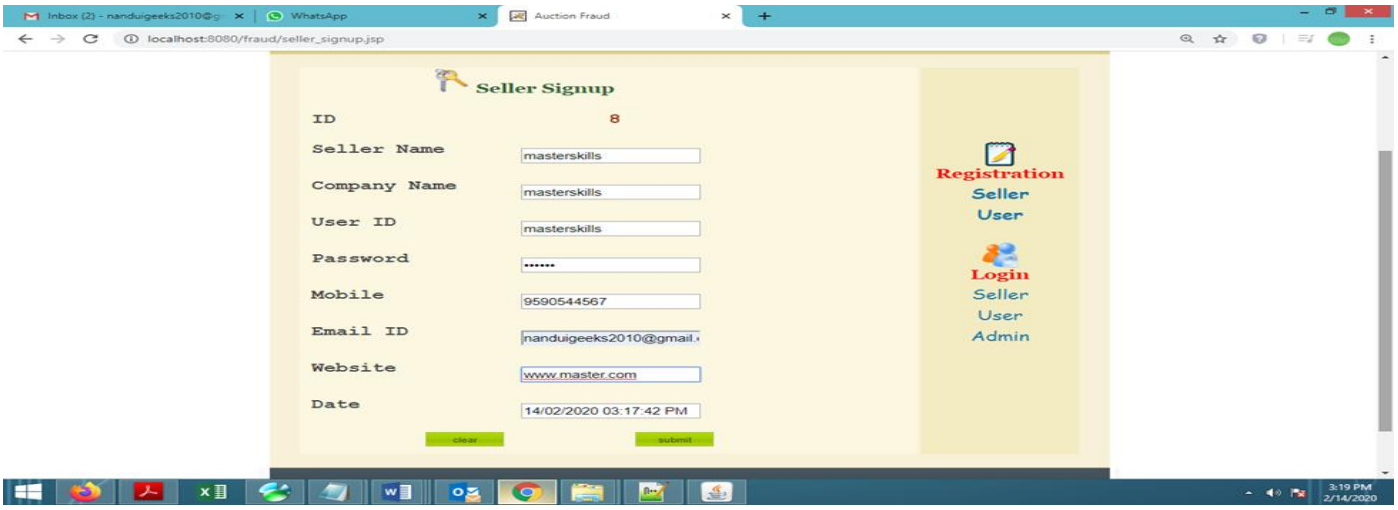
```
<%@page
import="com.oreilly.servlet.*,java.sql.*,java.lang.*,databaseconnection.*,java.tet
.SimpleDateFormat,java.util.*,java.io.*,javax.servlet.*, javax.servlet.http.*" %>
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<html>
```

```
`
<head>
<title>treasure warehouse</title>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">
<script type="text/javascript">
</script>
</head>
<body>
<%
Connection con=null;
PreparedStatement psmt1=null;
String a=request.getQueryString();
String tr="Registered";
try{
con=databasecon.getconnection();
psmt1=con.prepareStatement("update seller set authorize='"+tr+"'  where
uid='"+a+"'");
psmt1.executeUpdate();
response.sendRedirect("admin_seller.jsp");
}
catch(Exception ex)
{
out.println("Error in connection : "+ex);
}
%>
</body>
</html>
```

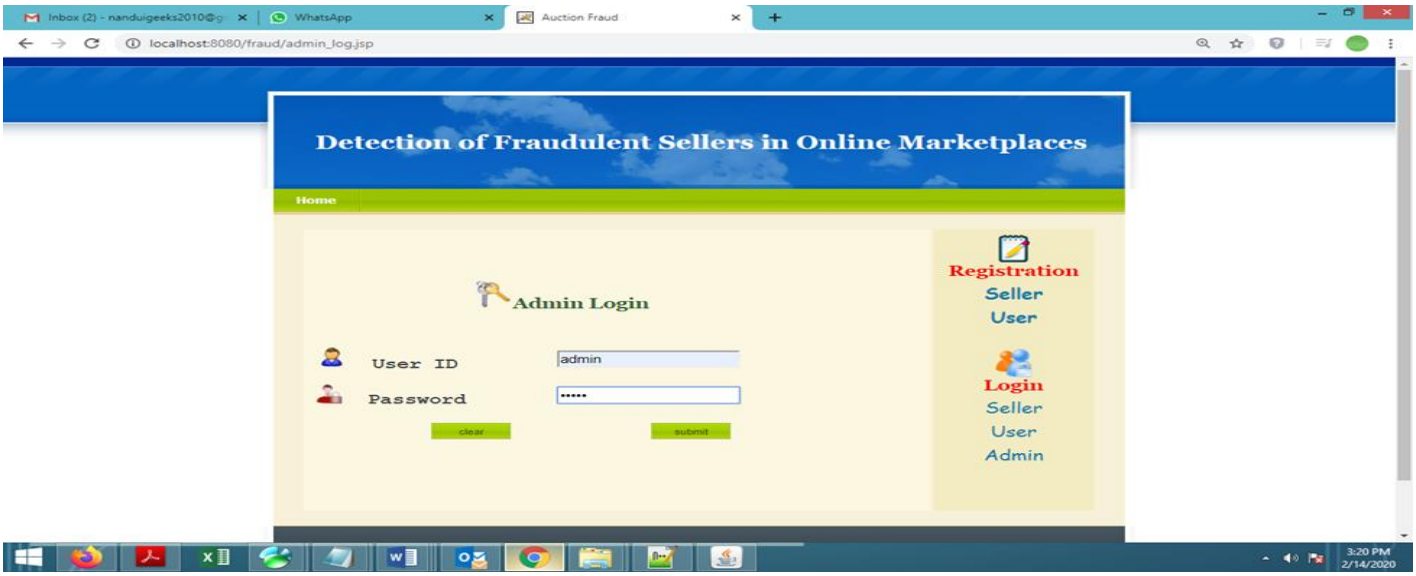
`

`

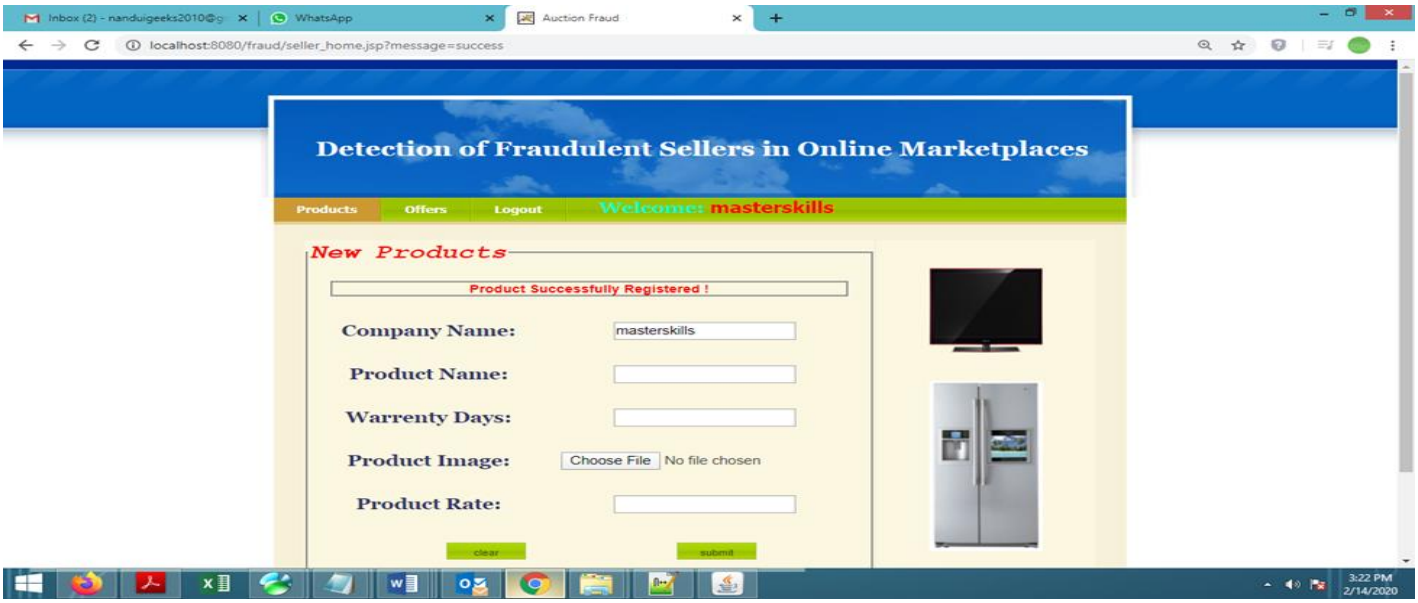
APPENDIX C: OUTPUT SCREENS



`

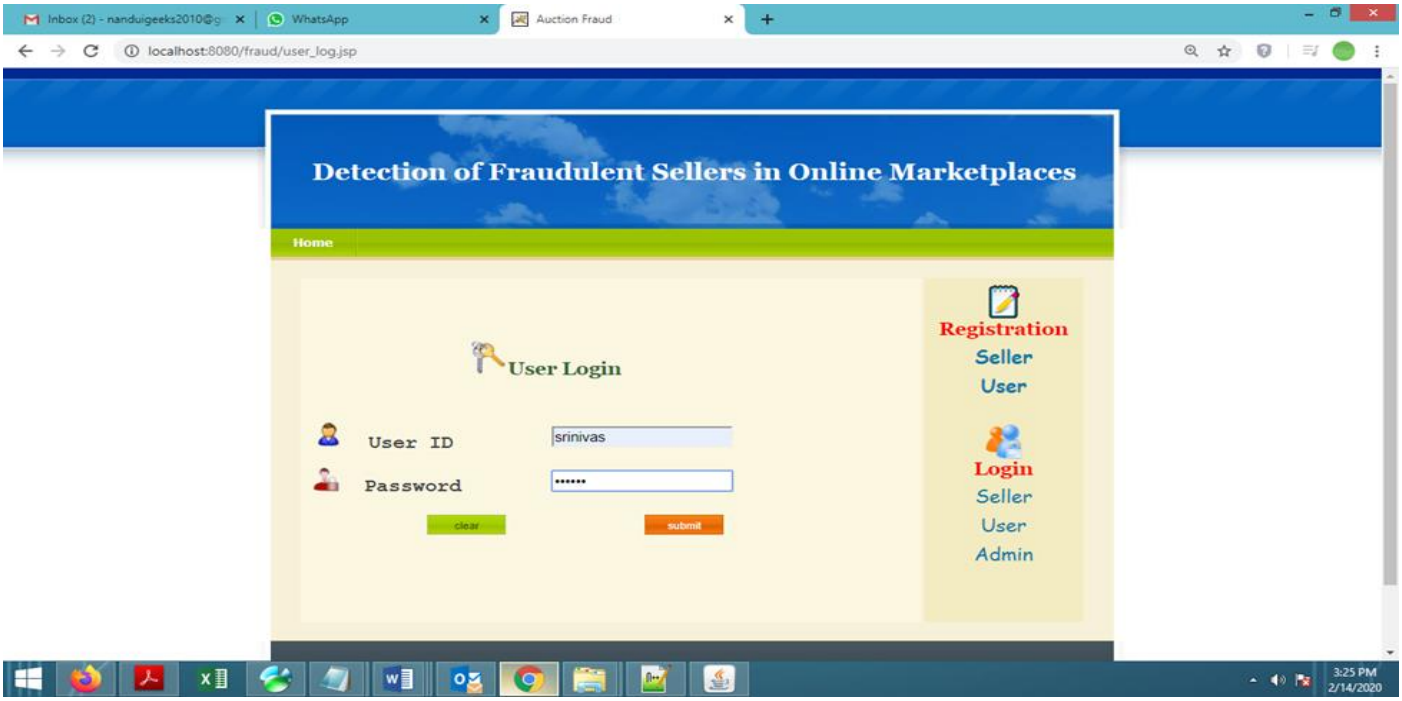

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/admin_log.jsp
Detection of Fraudulent Sellers in Online Marketplaces
Home
Admin Login
User ID
admin
Password
clear
submit
Registration
Seller
User
Login
Seller
User
Admin
3:20 PM
2/14/2020

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/seller_home.jsp
Detection of Fraudulent Sellers in Online Marketplaces
Products
Offers
Logout
Welcome: masterskills
New Products
Company Name:
masterskills
Product Name:
Warrenty Days:
Product Image:
Choose File
No file chosen
Product Rate:
clear
submit
3:21 PM
2/14/2020

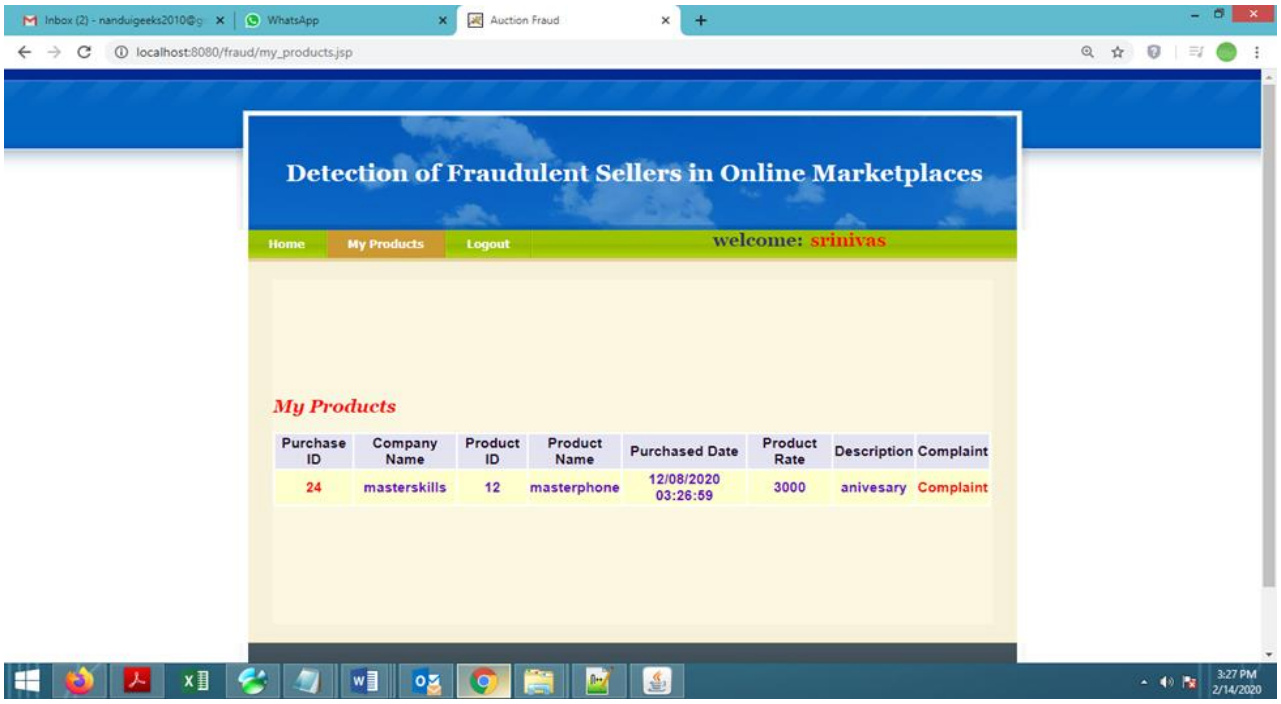

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/seller_home.jsp?message=success
Detection of Fraudulent Sellers in Online Marketplaces
Products
Offers
Logout
Welcome: masterskills
New Products
Product Successfully Registered !
Company Name:
masterskills
Product Name:
Warrenty Days:
Product Image:
Choose File
No file chosen
Product Rate:
clear
submit
3:22 PM
2/14/2020

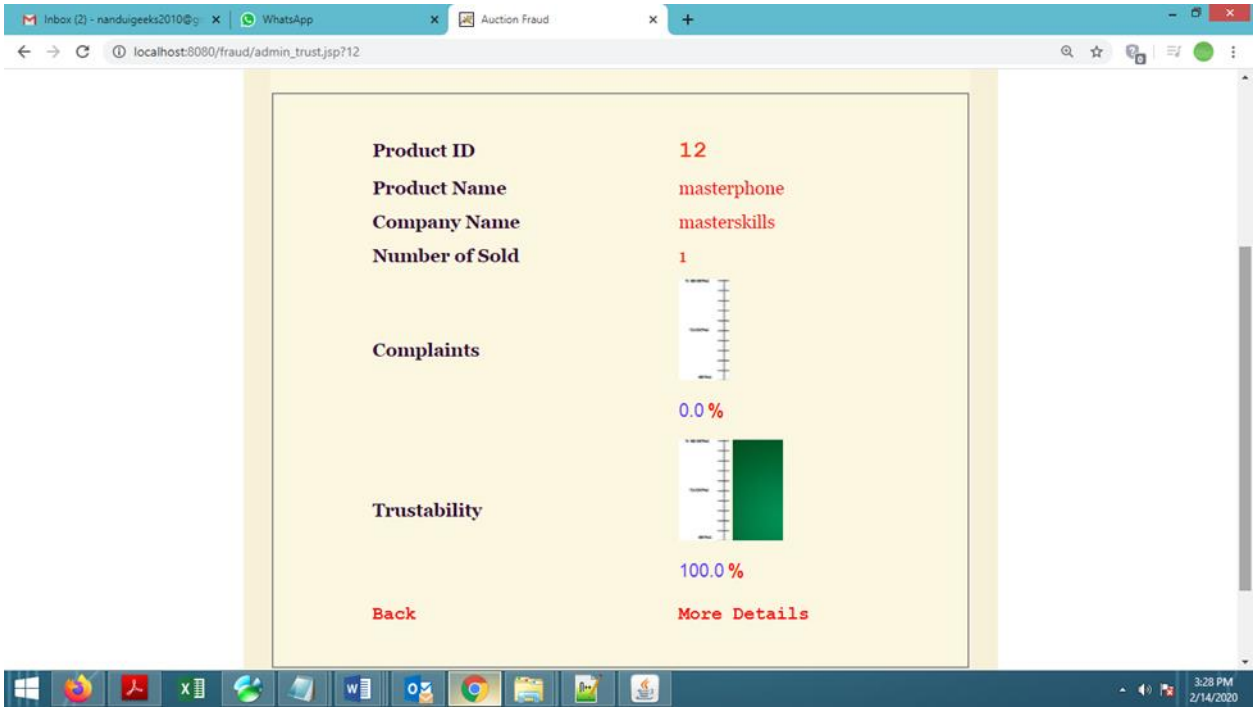
`

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/off_search.jsp?12
Products
Offers
Logout
Offers
Product ID: 12
Company Name: masterskills
Product Name: masterphone
Features : 180
Product Rate: 5000
Offer Rate: 3000
Offer Description: anivesary
Status: Registered
Trust: Trusted
Product Image:
Trustability
100 %
3:26 PM
2/14/2020

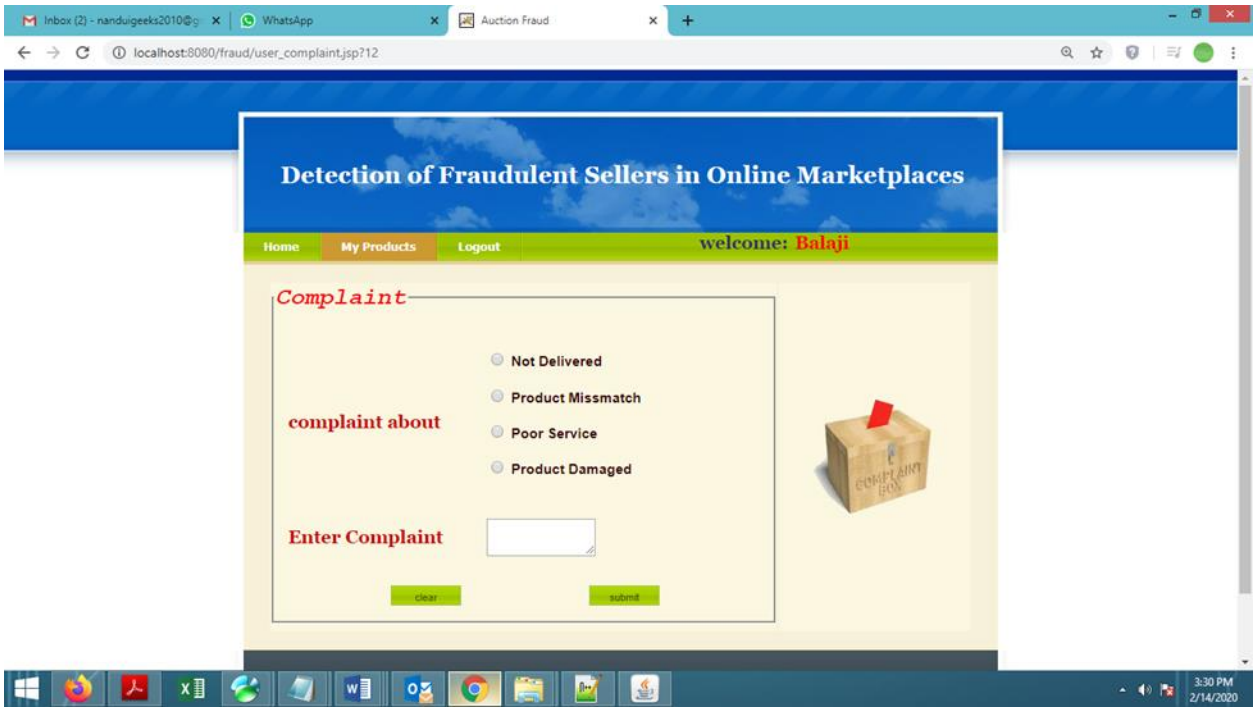

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/user_log.jsp
Detection of Fraudulent Sellers in Online Marketplaces
Home
User Login
User ID
srinivas
Password
clear
submit
Registration
Seller
User
Login
Seller
User
Admin
3:25 PM
2/14/2020

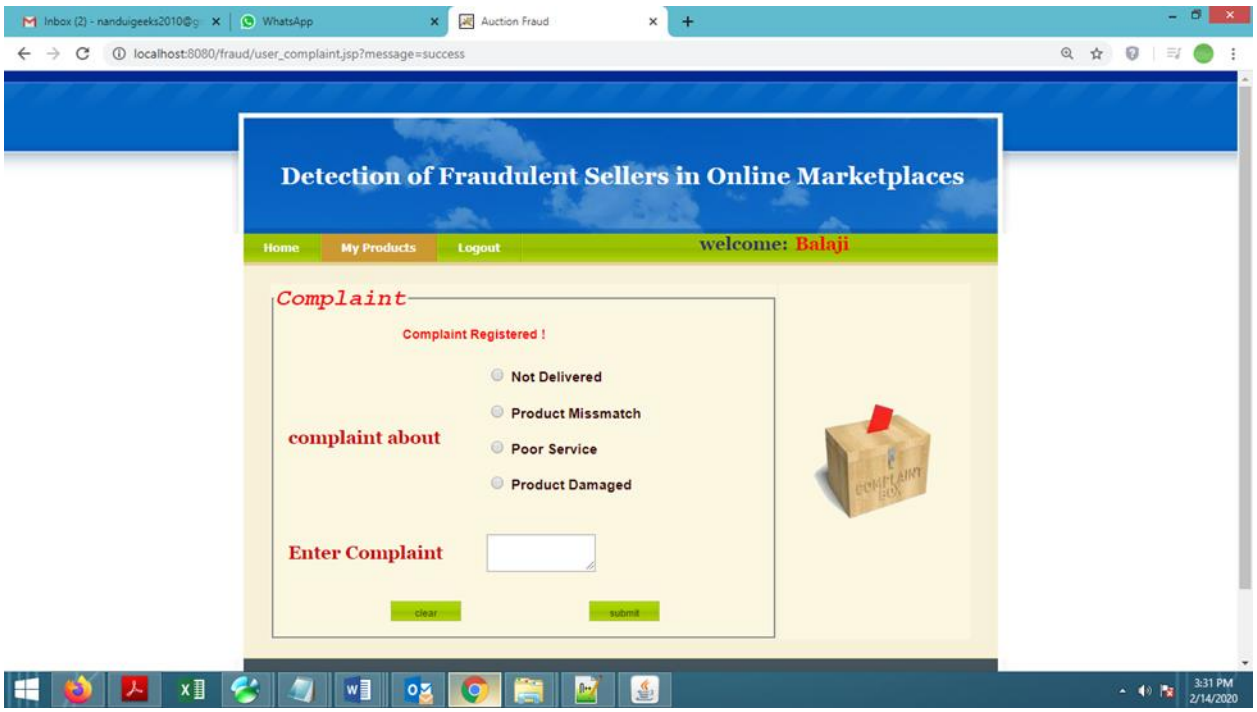
`


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/my_products.jsp
Detection of Fraudulent Sellers in Online Marketplaces
Home
My Products
Logout
welcome: srinivas
My Products
Purchase ID
Company Name
Product ID
Product Name
Purchased Date
Product Rate
Description
Complaint
24
masterskills
12
masterphone
12/08/2020 03:26:59
3000
anivesary
Complaint
3:27 PM
2/14/2020


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/admin_trust.jsp?12
Product ID
12
Product Name
masterphone
Company Name
masterskills
Number of Sold
1
Complaints
0.0 %
Trustability
100.0 %
Back
More Details
3:28 PM
2/14/2020

`


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/user_complaint.jsp?12
Detection of Fraudulent Sellers in Online Marketplaces
Home
My Products
Logout
welcome: Balaji
Complaint
complaint about
Not Delivered
Product Missmatch
Poor Service
Product Damaged
Enter Complaint
clear
submit
3:30 PM
2/14/2020


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/user_complaint.jsp?message=success
Detection of Fraudulent Sellers in Online Marketplaces
Home
My Products
Logout
welcome: Balaji
Complaint
Complaint Registered !
complaint about
Not Delivered
Product Missmatch
Poor Service
Product Damaged
Enter Complaint
clear
submit
3:31 PM
2/14/2020

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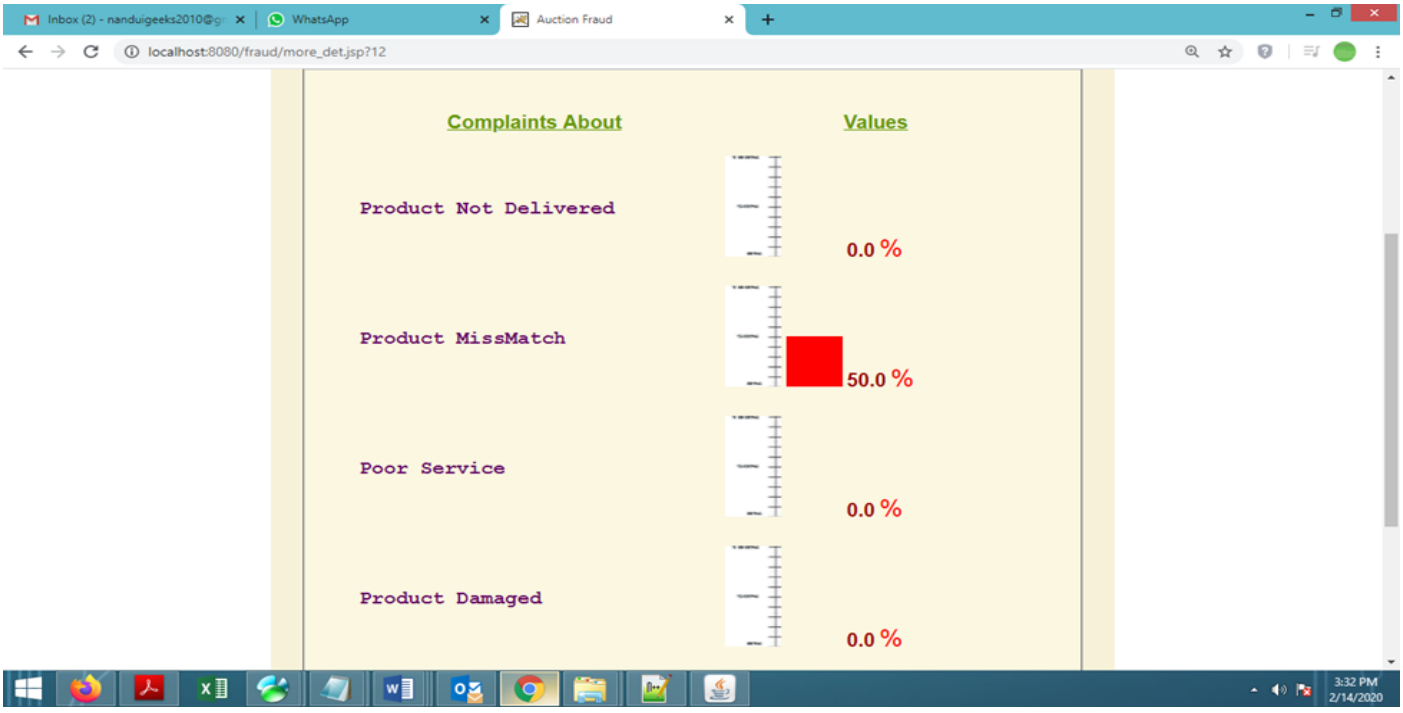
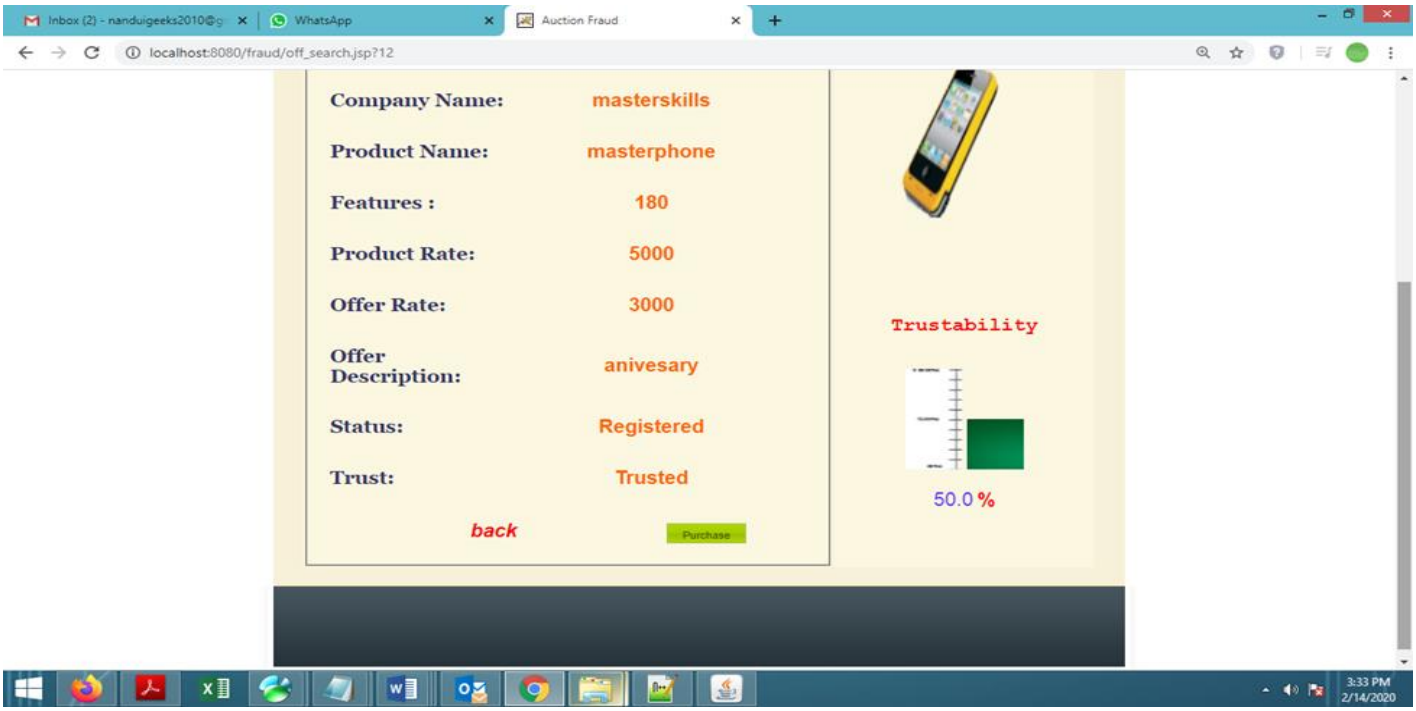
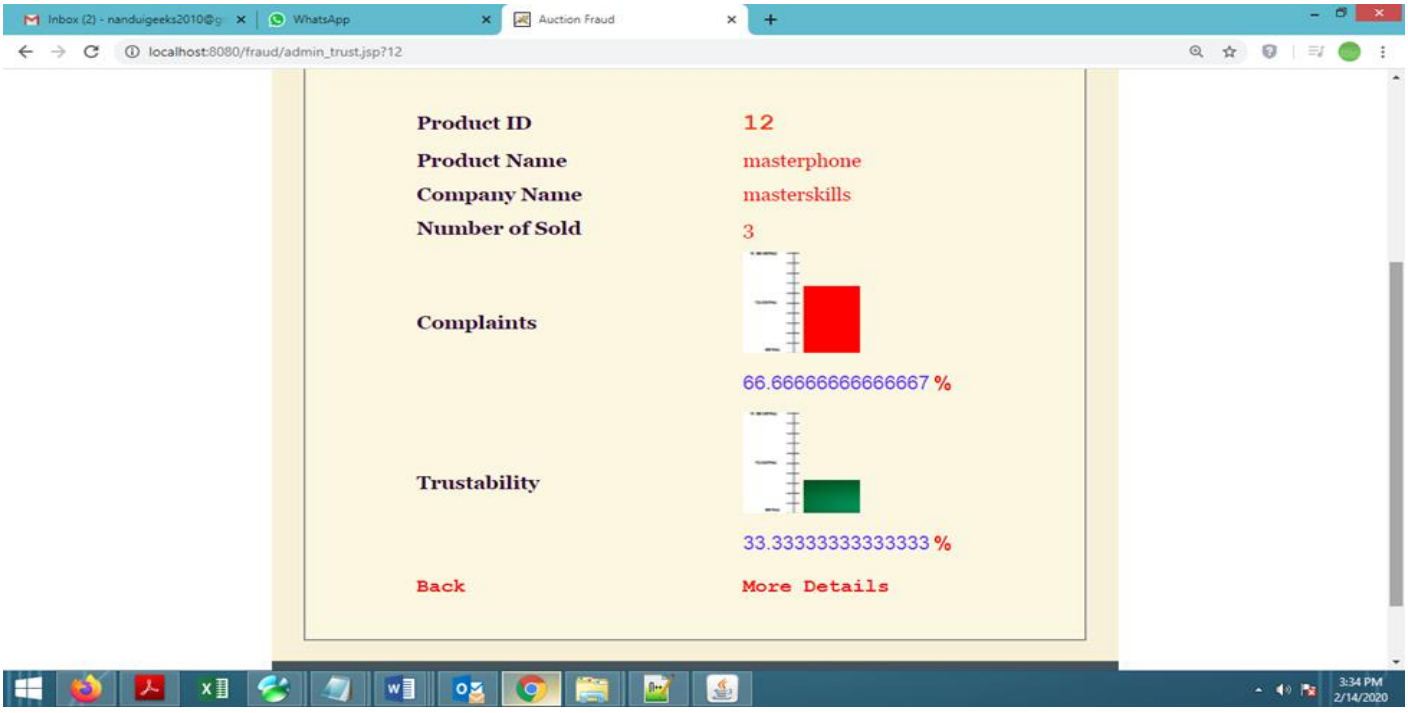
Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/admin_trust.jsp?12
Product ID
12
Product Name
masterphone
Company Name
masterskills
Number of Sold
2
Complaints
50.0 %
Trustability
50.0 %
Back
More Details
3:32 PM
2/14/2020

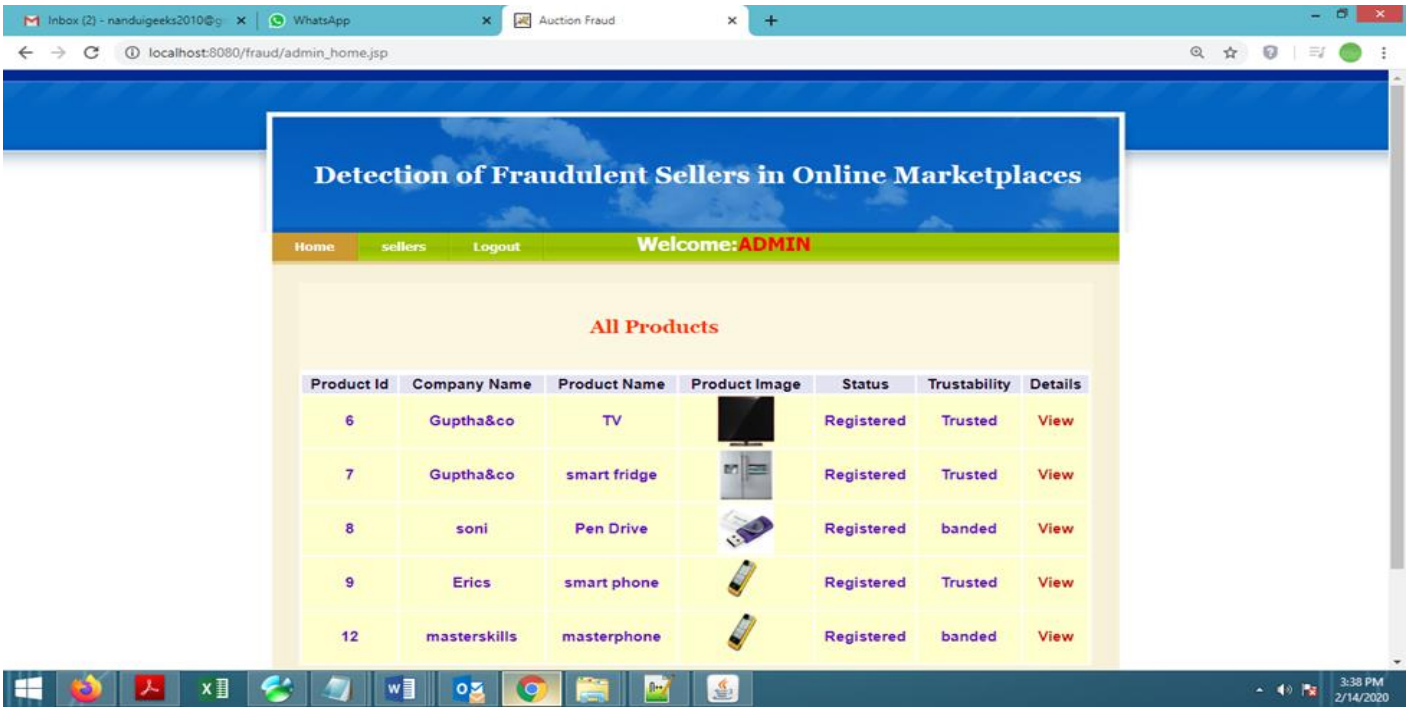

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/more_det.jsp?12
Complaints About
Values
Product Not Delivered
0.0 %
Product MissMatch
50.0 %
Poor Service
0.0 %
Product Damaged
0.0 %
3:32 PM
2/14/2020

`


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/off_search.jsp?12
Company Name: masterskills
Product Name: masterphone
Features : 180
Product Rate: 5000
Offer Rate: 3000
Offer Description: anivesary
Status: Registered
Trust: Trusted
back
Purchase
Trustability
50.0 %
3:33 PM
2/14/2020


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/admin_trust.jsp?12
Product ID 12
Product Name masterphone
Company Name masterskills
Number of Sold 3
Complaints
66.66666666666667 %
Trustability
33.33333333333333 %
Back
More Details
3:34 PM
2/14/2020

`


Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/admin_home.jsp
Detection of Fraudulent Sellers in Online Marketplaces
Home
sellers
Logout
Welcome:ADMIN
All Products
Product Id | Company Name | Product Name | Product Image | Status | Trustability | Details
6 | Guptha&co | TV | Registered | Trusted | View
7 | Guptha&co | smart fridge | Registered | Trusted | View
8 | soni | Pen Drive | Registered | banded | View
9 | Erics | smart phone | Registered | Trusted | View
12 | masterskills | masterphone | Registered | banded | View
3:38 PM
2/14/2020

Inbox (2) - nanduigeeks2010@g
WhatsApp
Auction Fraud
localhost:8080/fraud/user_home.jsp
Detection of Fraudulent Sellers in Online Marketplaces
Home
My Products
Logout
welcome: srinivas
Search offers
Product Id | Company Name | Product Name | Product Image | Status | Offers
6 | Guptha&co | TV | Registered | Offers
7 | Guptha&co | smart fridge | Registered | Offers
9 | Erics | smart phone | Registered | Offers
3:39 PM
2/14/2020